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MONDAY — JULY 12

DAY 1 TALK SUMMARIES

SCREENING AND VALIDATION OF CYANOBACTERIA METAL RESPONSE GENES ANNOTATED THROUGH METAGENOMIC DATA MINING

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Cyanobacteria or the blue-green algae are found in diverse and extreme environments from fresh water to marine or from damp soil to mine tailings. They are found thriving in various copper, gold and silver tailings storage facilities (TSF) withstanding extreme pH and temperatures. Cyanobacteria were isolated from water samples collected from Philex and Balatoc copper mine tailings in Benguet Province, Philippines. They were purified in BG-11 agar plates and cultivated in liquid BG-11 medium. Morphological characterization and molecular identification using the 16S rRNA gene revealed two distinct filamentous, Leptolyngbya sp. and Nodosilinea sp., and one unicellular Synechococcus sp. from the Order Synechococcales, and another filamentous cyanobacterium Phormidium sp. from the Order Oscillatoriales. The metadata processed using the cyanobacterial custom database of the metagenomic classifier CLARK v1.2 revealed a dominant cyanobacterium, Leptolyngbya sp., present across all sampling points of the Philex and Balatoc mine tailings. It is interesting to note that Nodosilinea sp. was not detected in the metagenomics analysis but was isolated in culture. It may be a unique Nodosilinean species that is why its sequence as obtained by 16S rRNA gene sequencing is not yet bound in any of the databases that were searched. This is the first report on the presence of Nodosilinea sp. in Benguet Province.
All four cyanobacterial isolates screened for the presence of metal response genes, namely, CopR, CsoR, CusA, CusB, CutA, Copres, CusR, Ca, SmtA, SmtB-trans, SmtB-HTH and Zn/Cd/Pb positively amplified the expected length of each metal response gene amplicon. Previously, the genes for the transcriptional activator protein (CusR), copper-sensing transcriptional repressor (CsoR) and copper-resistance protein A (CopresA) were only reported in sensing copper and silver concentrations in E. coli, Gram-positive bacteria and Acinetobacter, respectively. This study is the first to report the presence of these genes and their annotated copper-metal proteins in cyanobacteria isolated from copper mine tailings, particularly in the filamentous Leptolyngbya sp.

Comparing bioinformatic ASV-level pipelines for microbial V3-V4 amplicon 16S rRNA sequencing.

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Currently, there are several pipelines for analysis of variable fragments of 16S rRNA genes (Dada2, Deblur, Usearch, Mothur), however, these pipelines often give very different results for the same microbial communities. Our study aimed to compare the influence of different parameters of reads filtering and processing in Qiime2-Deblur and Dada2 pipelines on the OTU clustering results.
We compared two ASV-level bioinformatic pipelines (Dada2 and Qiime2-Deblur) for the analysis of V3-V4 region of 16S rRNA genes sequence data on a mock community samples as well as on the throat samples (N = 800), collected from the patients with respiratory diseases and healthy people. Given the great sensitivity and flexibility of the Dada2 algorithm, we tested the effects of different parameters (trimming, minQ filtering, paired reads merging, chimera detection and filtering) on the accuracy of taxonomic identification, as well as on the quantitative ratio of identified taxa in samples and mock controls.
We assessed degree of consensus of the different pipelines and workflows. To compare pipelines outputs, ASVs from different workflows were clustered into 97% OTUs, 99% OTUs and 100% ZOTUs with usearch-cluster and usearch-unoise3 algorithms (Edgar R.S, 2013; 2017; 2018). Our data were consistent with the results of the previous Dada2 workflows assessment, according to which, additional quality filtering in Dada2 shifted the taxa composition profile (Prodan A. et al, 2020). According to our data, chimera filtering parameters (minFoldParentOverAbindance = 15) and data filtering by quality (minQ =20) have the most significant effect on the composition profile and a risk of data loss

Genomic analysis testifies the important role of mobile elements in pathogenicity of Curtobacterium flaccumfaciens pv. flaccumfaciens

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Anna Lukianova (Shemyakin-Ovchinnikov Institute of Bioorganic Chemistry)
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A pathovar flaccumfaciens of gram-positive bacterium Curtobacterium flaccumfaciens (Cff) is the causal agent of bacterial wilt of Leguminosae plants, a disease that reduce the yields of many economically important crop around the globe. Previous genomic studies demonstrated the differences between pathogenic and not-pathogenic probably related to the genes encoding pectate lyase, serine proteases and cellulases. The main goal of the present research was to clarify the list of the genes responsible for the virulence of Cff and their origin. The emergence of an enormous quantity of new genomic data allows us to solve current problems in the taxonomic and evolutionary relationships of new species assigned to genus Curtobacterium.
To clarify the relations and check the taxonomic assignment within the genus Curtobacterium, we performed complex studies comprising the intergenomic comparisons and phylogenetic analysis employing different sets of genes. 16S rRNA phylogeny does not allow to differentiate the Curtobacterium strains into the species but discriminates them from other related genera. 23S rRNA analysis can resolve the distinction of different species, but still was not able to distinguish all the analyzed strains. The phylogenetic analysis with 91 house-keeping genes and their concatenated sequences has separated groups of strains assumed as potential distinct species, but at the same time demonstrated the presence of a continuum of genetic variability of Curtobacterium and several cases of strain misclassification. Besides, the phylogeny of 570 concatenated homologous protein sequences showed alterations in the topology of the tree.
The previous studies testified the role of plasmids in the pathogenicity of a number of bacteria. Moreover, some not-pathogenic bacteria were shown to develop such properties after they acquired such plasmids. We analyzed the genomes and plasmids of pathogenic and not-pathogenic Curtobacterium strains and found possible pathogenicity islands in the examined bacterial genomes and plasmids. The GenBank search demonstrated the presence of homologs of identified virulence genes in various bacteria including belonging to the taxonomically distant groups. Phylogenetic reconstruction has made it possible to propose ways of acquiring these genes. Analysis of plasmid genes demonstrated the evolutionary story of formation of plasmids related to the Curtobacterium virulence.
This research was supported by Russian Science Foundation grant #21-16-00047

Insights into the cyclooxygenase phylogeny

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Cyclooxygenases (COXs) and lipoxygenases (LOXs) are the most important oxylipin-forming enzymes in humans and other vertebrates. They have great medical significance as drug targets to treat inflammation, pain and fever. Fungi and plants also have cyclooxygenase homologs that participate in oxylipin biosynthesis and provide immunity signalling, quorum sensing and sexual reproduction. COX- and LOX-like enzymes are also the backbone of the oxylipin biosynthetic pathway in these organisms. COX- and LOX-like enzymes are also described in some bacteria [https://doi.org/10.1134/S0006297920090059, https://dx.doi.org/10.1074/jbc.M114.555904]. Similar functions of COX- and LOX-like enzymes in different kingdoms of life and their evident homology raise a question about their origin and evolution.

Recently, we published a paper on LOX evolution [https://doi.org/10.1134/S0006297920090059], where found both a statistical and an evolutionary association between LOXs and multicellularity, as well as a 'minor' association between LOXs and cross-kingdom virulence. We assumed that LOXs could have originated in multicellular bacteria and be borrowed by different groups of eukaryotes by horizontal gene transfer to provide cell-to-cell signalling.

Here, we present a follow-up of that study aimed to elucidate the second core component of all oxylipin biosynthetic pathways – COX-like enzymes. Such a research has already been published by Gupta and Selinsky [https://doi.org/10.1016/j.bbamem.2014.09.011], but includes a small sequence sample and the resulting phylogeny of bacteria COX-like enzymes seems controversial.

We have performed a phylogenetic research for animal COXs, plant alpha-dioxygenases (αDOXs) and fungal Psi-producing oxygenases (PPOs) with a larger amount of bacterial sequence data. We have revealed that plant αDOXs and fungal PPOs have originated independently of animal COXs from bacterial peroxidase-like enzymes with unknown function. Animal COXs appeared to be closely related to some groups of bacterial enzymes, including the single described cyanobacterial COX-like enzyme [https://dx.doi.org/10.1074/jbc.M114.555904]. Combining available PDB data with sequence alignments, we have found that COXs, αDOXs and PPOs share common catalytic residues (that are also shared with peroxidases and catalases), but their substrate-binding residues developed independently. It confirms our assumption that COXs, αDOXs and PPOs originated independently from peroxidase-like enzymes despite very similar biochemical and biological function.

Substrate-binding facility appeared to be homologous only in animal COXs and the cluster of bacterial COX-like enzymes closely related to them. A detailed statistical analysis of this cluster revealed the highest abundance of COX-like enzymes in multicellular bacteria (Cyanobacteria and Actinobacteria), similar to observed peaks of LOX abundance in multicellular Myxococcales and Cyanobacteria. It probably reflects that bacterial COX-like enzymes are also associated with cell-to-cell signalling and forming complex structures and raises a possibility that they could also originate and spread across the life to support multicellularity. These findings support the hypothesis that oxylipins have initially originated to provide cell-to-cell signalling even at the 'bacterial' stage of their evolution [https://doi.org/10.1134/S0006297920090059].

NGS data for natural GMO discovery and investigation

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Agrobacterium-mediated transformation is the most common method for obtaining commercial lines of transgenic plants. Despite the fact that this method is based on a natural vector system, genetically engineered organisms (GMOs) are constantly attacked as organisms alien to nature. An important argument in support of GMOs was the discovery of plants transformed by agrobacteria in nature millions of years ago [1]. Representatives of the genus Nicotiana were the first discovered examples of natural GMOs (n GMOs) [2]. DNA sequences in their genomes, homologous to agrobacterial ones, are called cellular T-DNA (cT-DNA). Thirty years later, nGMOs have been described within the genera Linaria and Ipomoea [3-4]. And only the appearance of extensive data of sequenced plant genomes made it possible to adequately assess the scope of the phenomenon [5-6]. Our 2019-2021 studies increased the number of known species of natural GMOs by an order of magnitude, as well as trace the phylogenetic relationships of sequences homologous to agrobacterial ones in plant genomes. Based on the analysis of NGS data, it was possible to describe cTDNA in representatives of more than 30 genera of angiosperms, to clarify the peculiarities of cTDNA organization in representatives of the genus Ipomoea, to study cTDNA polymorphism in representatives of the genera Nicotiana and Camellia.

Analysis of cT-DNA sequences allows us to draw preliminary conclusions about the relationship of bacterial strains that participated in the transformation of plants millions of years ago, to identify the most conservative genes and intact ORFs, as well as the patterns of their distribution among natural GMOs. Based on the accumulated data, it is already possible to draw a conclusion about the predominance of opine synthesis genes over other groups of genes in cT-DNA, as well as a higher percentage of intact ORFs among them [7].

All the issues mentioned above will be discussed in the presentation together with the great social significance of the result of this work. By now, it has become clear that humanity throughout its history has come across transgenic plants, using them for food, for the treatment of diseases, for technical or decorative purposes.

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The search of somatic embryogenesis regulators in plants

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Somatic embryogenesis (SE) is the way of regeneration of plant, during which the vegetative cells form embryos, which potentially can develop into the new plant. This process is used in the plant biotechnology for genome modification and plant breeding. The ability of plant to form somatic embryos in vitro depends on cultivation conditions and also on the plant genotype. Therefore, the investigation of genes regulating this process can help to improve the SE protocols. We search for SE regulators among the genes from WUSCHEL-RELATED HOMEOBOX (WOX) and NUCLEAR FACTOR Y (NF-Y) families, encoding transcription factors, as well as among the genes from CLAVATA3/ EMBRYO SURROUNDING REGION (CLE) family, encoding peptide hormones, using Medicago truncatula (barrel medic) as model object.

We found that MtWOX9-1 and STENOFOLIA genes from WOX family are able to stimulate SE in M. truncatula when overexpressed. Edited plants with loss of MtWOX9-1 function are sterile, which supports the role of this gene in the embryogenesis. Transcriptomic analysis of calli with MtWOX9-1 overexpresion, as well as analysis of its coexpressed transcripts, uncovered a lot of potential targets of this transcription factor, which include the genes from NF-Y and CLE families.

We also found one possible suppressor of SE among CLE genes, which can be used as potential target for editing to obtain the plants with high capacity for regeneration.

The research was made with support of the Ministry of Science and Higher Education of the Russian Federation in accordance with agreement № 075-15-2020-922 date 16.11.2020 on providing a grant in the form of subsidies from the Federal budget of Russian Federation. The grant was provided for state support for the creation and development of a World-class Scientific Center “Agrotechnologies for the Future”.

Testing the EXONtools pipeline: pseudoreference development and hybridization bait design for exon capture experiments

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The EXONtools pipeline was proposed for exon capture sequencing data analysis on non-model species. It was initially presented at SMBE2017 and included three major stages: (1) de novo transcriptome assembly and annotation; (2) exon hybridization bait development; and (3) exon-intron loci assembly with ortholog alignment and SNP calling. Thereafter the EXONtools algorithms and the program code have been sufficiently improved and approbated in several population genomic studies. The pipeline was also adjusted to work with amplicon, RAD-seq, and UCE data. Each exon capture sequencing analysis on non-model species requires some sort of a reference, such as a well-annotated genome or transcriptome, to be used as a template for hybridization baits. These baits should be designed to have a high enough nucleotide sequence similarity with their target regions that allow for successful gene capture (i.e., hybridization) across all individuals used in a population or phylogenomic study. The EXONtools pipeline uses a consensus assembly approach to produce high-quality transcripts and exon sequences that correspond to protein-coding regions, and therefore, can be used as references for exon capture bait design. I call such dataset “pseudoreference” because it usually does not refer to any particular individual but represents a combination of various references. The major problem in such an approach is the presence of different artifacts in a final consensus assembly including (a) incomplete contigs having long gaps (scaffolds); redundant sequences representing the same transcript (isotigs); (b) transcript isoforms produced by alternative splicing events (isoforms); (c) paralogous transcripts resulting from gene duplications (paralogs); and (d) chimeric contigs (misassemblies). Leaving these artifacts untreated will cause a strong decrease in bait hybridization efficacy, as well as an appearance of false-positive SNP calls in resulting exon capture data. Thus, in the current study, I examine each of these constraints and propose solutions to how they can be unraveled and processed by using the EXONtools pipeline. I tested the performance and quality of de novo transcriptome assemblies produced by three popular assembly programs incorporated in the EXONtools pipeline. The final dataset (pseudoreference) included >42K annotated contigs with predicted CDS regions, including >25K unique annotations and >130K unique exon templates for the non-model species Stenogobius sp. The Stenogobius pseudoreference was used to design hybridization baits for exon capture experiments on amphidromous gobies (Teleostei: Gobiiformes). The proportions of highly divergent and highly conserved exons in the newly-developed Stenogobius pseudoreference were estimated by measuring p-distances between exon sequences and the orthologous regions in model fish species. The current study demonstrates how the EXONtools pipeline can be applied for any non-model species of interest to produce hybridization baits with a high capture efficacy. As a universal tool for the development of numerous genome-wide molecular markers, the EXONtools should boost the performance of phylogenomic and population genomic studies of non-model species to a much greater level.

The reported study was funded by RFBR, project number 19-37-51019 (pipeline development and software testing), and Ministry of Science and Higher Education of the Russian Federation, project number FZNS-2021-0011 (experimental data tests and pipeline approbation).

Genome-wide Maps of Background Allelic Dosage Reconstructed from Variant Calls with Bayesian Change-point Identification

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Rapidly advancing DNA sequencing technologies provide consistent improvement in detection of global and local Copy Number Variants (CNVs), making it possible to infer copy numbers of particular alleles. However, it is still necessary to have experimental whole-genome data, either results of deep sequencing data or allele-specific genotyping by SNP microarrays. Additionally, long reads sequencing data is required for reliable reconstruction of individual haplotypes.
Genome-wide copy-number profiles are very important for computational discovery of allele-specific gene events involved in transcription activity regulation (transcription factor binding or chromatin accessibility modifications) and broader in gene expression. Quite often neither genome copy-number profiles nor the experimental data are available for copy number profile reconstruction for a particular cell type or tissue sample under study. Yet in many types of analysis only the relative dosage of alternative alleles is sufficient rather than exact copy numbers of genome segments.
Here we present a novel algorithm for efficient inference of full-genome maps describing relative background allelic dosage. The algorithm constructs such
maps directly from variant calls and the results are consistent between target sequencing (such as ChIP-Seq or DNase-Seq) and whole-genome sequencing data.
We show how the algorithm can be used in massive analysis of allele-specific transcription factor binding, where background allelic imbalance explains the overdispersion of read coverage distribution.

Molecular modeling predicts a gene function, reveals evolutionary constraints and opportunities of the gene, and thinks about the true amino acid substitution matrix

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Molecular modeling is a powerful tool for understanding the details of gene function, the meaning of the amino acid substitutions, and a way to approach some fundamental problem of computational evolutionary biology. Here we present the story where the molecular modeling helps solve one of the longest-lasting questions in the plant-microbe interaction - the genetic control of the "afghan" phenotype in pea-rhizobia symbiosis. The main problem was to test candidate pea receptor genes to potentially be responsible for the "afghan" phenotype, i.e. the differential interaction with the acetylated and non-acetylated forms of the rhizobial Nod factor. Molecular modeling of the receptor dimer and the docking of both Nod factor versions gave the highly probable answer that it was the LykX gene (Solovev et al., 2021). The evolutionary expansion of this approach on another receptor dimer allowed us for the first-time visualization of the most intimate details of the evolutionary process in the contact surface between subunits of the dimer with the docked Nod-factor. The "projecting" algorithm superimposes population polymorphisms of subunits on their contact surface together with the Nod-factor docked, which highlighted the essential area of the signal system in the context of interaction between receptor and ligand (Igolkina et al., 2018). Now, using molecular modeling, we can estimate the effect of any amino acid substitution on the efficiency of the interaction between the pea receptor and rhizobial signal molecule. But this field has a rather deep extension. Indeed, if the molecular modeling makes it possible to calculate the effect of any substitution in any amino acid position in the protein, we could try to build the "true" amino acid substitution matrix, which has to be: 1) specific for each particular protein; 2) and specific for each ligand/substrate; 3) of a complex structure, for example, constructed as the stack of partly filled square matrices (one per amino acid position). But if we take into consideration the epistatic interactions between amino acid positions we must admit that this model of the "true" matrix is just a modest approximation to hardly achievable ideal.

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PlasPred – Computational Prediction Tool for Identifying Plastic-Degrading Microbes

Hemalatha N (Associate Professor)
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Plastic pollution has become one of the most stressing environmental issue, as rapidly increasing production of disposable plastic products overwhelms the world's ability to deal with them. A life without plastic is unimaginable. Plastics revolutionized areas such as medical field, space, transportation etc. whereas on other side it has created a major environmental pollution. Hence, it is practically impossible to avoid plastic completely from our daily life. Only solution to control this pollution is degrading these products rather than throwing it into the surroundings.
Plastic degradation is required for the successful management of sustainable development. Biological process plays an important role in removing the contaminants and also taking the benefit of microorganisms who are highly versatile to degrade these plastic components. The biopolymer microbes responsible for degrading plastics are a boon to the society where plastic is a severe problem.
A proper review of previous articles of degrading microbes were conducted and this helped us create a database of biodegradable and non-biodegradable microbes. In this work we have developed a computational prediction model with WEKA, a machine learning tool with various classifiers which will help in predicting whether a microbe has the degradation property or not. Plastic degrading protein sequences belonging to the alkB and CYP153 genes were cumulated from databases such as NCBI and UniprotKB. Around nine thousand positive protein sequences and six thousand negative sequences were obtained. For training datasets, we divided the datasets in the ratio 60:40. In this work, we have used six different features namely amino acid counts, dipeptide counts, amino acid ratio, hydrophobicity, hydrophilicity, acidity and basicity of microbial protein sequences.
Count of each amino acid in a sequence was divided by the length of protein sequence in the amino acid ratio for which feature dimension was 20. In the dipeptide count, count of the occurrence of all dipeptide in the protein sequence was taken and dimension was of size 400. In the physiochemical properties that included hydrophobicity, hydrophilicity, acidity and basicity, feature dimensions were nine, six, two and three simultaneously. WEKA, an open-source machine learning software has been used in this work for practical implementation. Different classifiers used are SMO, MLP, J48 a decision tree classifier and Instance Based Learner (IBK). Machine learning models were created with all the four classifiers and six different features with our training set. It was found that all the four classifiers successfully classified the microbes with 100% accuracy for amino acid count feature where as for other features it was less than 95%. This feature was also considered the best as it had only a feature dimension of twenty and comparatively less than other features.
Plastic pollution is a major concern among the scientific community. There are chances that biologists may come across microbes which will have the potential of degradability. To conclude, this model can be used in cases where we want to test the degradability of such microbes which can degrade plastics.

Integrated Pipeline for Mining BGCs and RiPPs in Microbial Metagenomes

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Specialized metabolites (SMs) with pharmaceutical bioactivities are crucial to study to search for new antibacterial and anticancer agents from nature. One approach to discover new SMs is mining metagenomes of microbes living in uncharted habitats. To comprehensively investigate available microbial metagenomes, we assembled a pipeline to capture both the BGCs with similarity to known BGCs and BGCs with novel sequences. The sequence reads are filtered and assembled into contigs, then antiSMASH (Hidden Markov Model [HMM]-based) and DeepBGC (deep learning-based) algorithms are used to detect the BGCs in each of the assemblies. We processed 118 samples from 10 publicly available microbial metagenomic projects using our pipeline with a total size of ~1.5 Gbps. 1,316 BGCs were detected by antiSMASH in all the environments, while DeepBGC detected 124,118 BGCs. The BGCs identified by antiSMASH were of 31 different classes, with the largest three groups, NRPS, terpenes, and bacteriocins. Most DeepBGC detected BGCs were not assigned specific product classes (80% of the total clusters). Still, instead, they were mainly annotated based on their product activities, such as antibacterial and antifungal. Combining the two approaches captured more BGCs, including novel ones. Here, we focus on the RiPPs class (Ribosomally synthesized and post-translationally modified peptides). Eight different RiPP cluster types were detected by antiSMASH: bacteriocin, lanthipeptide, LAP, lasso peptide, RaS-RiPP, sactipeptide, TfuA-related, and thiopeptide. AntiSMASH seems to be more accurate in detecting and annotate the different RiPPs, particularly bacteriocins. DeepBGC was advantageous in predicting many more BGCs based on their activities. We are currently expressing selected BGCs to assess their activity in vitro. The two approaches complement each other and improved the power of detection of each BGC type beyond applying a single method. Our results are expected to be the basis for finding novel bioactive agents from microbes living in diverse ecosystems.

IDOPS, a profile HMM-based tool to detect pesticidal sequences and compare their genetic context

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Biopesticide based crop protection is challenged by insect resistance. Thus the constant expansion of available biopesticides is crucial for sustainable agriculture. Although Bacillus thuringiensis is the major agent for pesticide bioprospection, the number of bacteria species synthesizing proteins with biopesticidal potential is much higher. The Bacterial Pesticidal Protein Resource Center (BPPRC) offers a database of sequences for control of insect pests, grouped in structural classes. Here we present IDOPS, a tool to detect novel biopesticidal sequences and analyze them within their genetic environment. The backbone of IDOPS detection unit is a curated collection of high-quality hidden Markov models that is in accordance with the BPPRC nomenclature. IDOPS was positively benchmarked with BtToxin_Digger and Cry_Processor. In addition, a scan of the UniProtKB database with IDOPS models returned abundant new candidates to pesticidal proteins distributed in all of the structural groups. Gene expression depends on the genomic environment, therefore, IDOPS provides a comparative genomics module to investigate the genetic regions surrounding pesticidal genes. This feature enables the investigation of accessory elements and evolutionary traits relevant for optimal toxin expression and functional diversification. IDOPS contributes to study and expand our current arsenal of pesticidal proteins used for crop protection.

TUESDAY — JULY 13

DAY 2 TALK SUMMARIES

Near-saturating semi-automated inference of bacterial transcription factor binding sites for regulatory network analysis and producer strain construction

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In the present era of exponentially growing genomic databases, bacterial genome annotations mostly lack information about regulatory elements controlling genome expression: promoters, terminators, and transcription factor binding sites (TFBSs). In silico approaches exist for inference of all three types of regulatory elements, but TFBSs are arguably the most difficult to analyse. Due to the numbers and high variability of TFBSs, complete genome-scale TFBS analysis was impractical until recently.

We have previously presented the SigmoID v2 application, which contains the pipeline for the genome-wide inference of TF binding sites (TFBS) in bacterial genomes, as a tool for resolving regulation details for paralogous TFs [1, 2]. The pipeline exploits the idea of critical residue tags (CR tags) – a set of amino acid residues making direct contacts with DNA bases within TFBS) as the unique identifier of the TF-TFBS pair. Using de novo TFBS pipeline, one could infer operators of specific TFs, producing a set of testable regulated targets. An inverse problem is often encountered: the specific genes of interest are known and annotated in a genome, but their plausible regulators remain hidden. This problem is typical for biotechnology projects aiming at increasing the expression of a particular protein. In such a situation, looking for regulators of a single gene still requires checking all transcription factors present in the genome. Since the last stage of our pipeline relies on human decision on the best motif, complete analysis of TFBSs for all transcription factors is time-consuming.

Here we describe an approach for fully automated TFBSs analysis for almost all TFs encoded in a bacterial genome. Rather than deciding on the best motif for each TF, it adds all inferred motifs to the annotation. This allows the user to see all possible motifs in front of the target gene, perform a small-scale verification of these motifs and thus find possible regulators for the target genes much faster. We hope this approach could be useful for biologists designing strain modifications.

We have also introduced important changes to the TFBS inference pipeline to bring it closer to saturating whole-genome analysis. First, we have added analysis of TFs with two DNA-binding domains (typical for, e.g. the AraC family). Second, we now properly model variable linkers between the HTH recognition helix and additional DNA base contacting residues. Third, we have added models for non-HTH TFs. Fourth, we have used structure-based modelling to infer CR tags for TF families with no 3D structures available. These improvements allow SigmoID to analyse over 90% of TFs for Enterobacteriales and allow to consider such analysis as near-complete. CR tags for TF families present in other taxonomic groups could easily be added.

Experimental data confirming the efficiency of the targeted TFBS inference approach will be presented.

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Evolution of complexity: transcriptome age indexes analysis in a complex life cycle

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The rapid development of high-throughput sequencing technologies made the detailed description of the molecular basis of biology possible for the wide range of organisms. The vastly growing number of sequenced genomes and transcriptomes allows researchers to perform a comparative transcriptomic analysis both within and between species. Consequently, the questions about evolutionary novelties and conservative components are rising.
One of the most promising and powerful approaches to trace evolutionary innovations using data from genome projects is “phylostratigraphy” proposed in 2007 by Domazet-Lošo, Brajkovic´ and Tautz. Based on the identification of gene founder for lineage or family and punctuated emergence of protein families, the phylostratigraphy provides a statistical approach for the reconstruction of macroevolutionary trends. Given the information obtained, one can compare the contribution of different groups of genes with a common phylogenetic origin to the transcriptome. The higher the value of the transcriptome age index, the greater the contribution of the “new” genes. In theory, the described approaches can serve as a basis for reconstructing the sequences of evolutionary events that underlie the analyzed trait. Primarily this concerns the traits for which it is almost impossible to trace the evolutionary history using fossil data. The complication of the life cycle of parasitic flatworms is one of the remarkable examples.
Here, publicly available transcriptomic data were used for the identification of stage-specific molecular signatures in the complex life cycles of 6 parasitic flatworm species. For each of the signatures, an age index was determined using the results of phylostratigraphy analysis. Significant differences between the stages in terms of age indexes were revealed both in the analysis of complete transcriptomes and reduced datasets without relatively “new” genes. For example, the highest values of the indices are characteristic for the stage of the cycle, which is specific only for 1 digenean family and, presumably, appeared one of the last in the cycle. At the same time, it is difficult to identify the common sequence of the appearance of stages in the cycle of a hypothetical ancestor. The high transcriptome plasticity of stage-specific signatures and differences between species in the data available for various stages are among the main reasons.
However, it is important to note that each of the stages of cycle is characterized by many traits that have probably been evolved independently. Further studies are needed to answer the question of whether it is possible to distinguish signals from different traits in the age indices of transcriptomes. A remarkable example is a correlation between high values of the age index and the activity of processes associated with the nervous system. Complete and detailed reconstruction of the evolutionary sequence of life cycle complications using data on modern species remains a great challenge for both zoologists and bioinformatics.

MMseqs2 profile/profile: fast and ultra sensitive searches beyond the twilight zone

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Analyses of high-throughput sequencing studies are producing billions of novel, uncharacterized protein sequences of previously unstudied organisms. State-of-the-art sequence-to-sequence alignment methods are well-tuned to cope with the avalanche of data. However, they are limited by their sensitivity as they rely on existing homologous sequences in reference databases within the daylight-zone of detectability. The most sensitive alignment methods, HHblits and HHsearch, are based on HMM to HMM (profile-profile) alignments and are able to detect remote homologies across vast evolutionary distances, well into the midnight-zone. However, they are limited in their applicability since they take minutes to process a single query, despite large optimization efforts. Here, we propose MMseqs2 Profile/Profile, the first profile-profile alignment method to match the sensitivity of HHblits at a much higher runtime speed. We extend the fast SIMD-accelerated implementation of the Striped Smith-Waterman-Gotoh algorithm in MMseqs2 to support profile-profile alignments and introduce efficient workflows for reverse- and iterative-searches for the construction of extremely diverse multiple sequence alignments. At nearly 4100x the speed of HMMER3 while being 20% more sensitive. At over 60x the speed of HHblits, we match its sensitivity. Furthermore, we expect MMseqs2 Profile/Profile to scale well onto large query datasets, due to its efficient parallelization.

Optimal de novo assemblies for chloroplast genomes based on inverted repeats patterns

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It is well known that the chloroplast genomes possess a highly conserved circular and quadripartite structure with a pair of dispersed inverted repeat regions. Using this knowledge, we design a dedicated pattern-driven de novo assembler. Our pipeline requires short unpaired reads uniquely (distances provided by paired-reads are not needed), sequenced from both the plant and its chloroplasts. The first step consists in selecting the chloroplasts reads from the reads specific to plant. We use the k-mer frequencies to compute an estimated coverage on pre-assembled contigs in order to keep the ones with higher coverage because of the oversampling of the chloroplast genomes compared to the plant genome.
The first step outputs an assembly graph where each vertex/contig is provided with an estimated multiplicity number. In the sequel we use another graph, that we call explicit, where each vertex is duplicated according to its multiplicity number and to the two possible contigs orientations. The edges are duplicated respectively. In this graph the genome assembly corresponds to finding an elementary path. We formulate the dispersed repeats as linear constraints and we search the elementary path with Integer Linear Programming similarly to [1]. In our approach, inverted repeats correspond to occurrences of contigs paired with other occurrences of them but in reverse orientation. Their positions on the assembled sequence must satisfy nested-pairs pattern. The challenge was to model the above condition in terms of linear constraints. The objective of the linear program is to maximize the nested-pairs number. Thus we generalize a similar approach applied for RNA folding [2]. Indeed, in contrast to the later approach where the vertices correspond to bases with known sequence indices, in our case the positions of the contigs are variables. Furthermore, we design an algorithm with linear complexity for generating multiple optimal solutions. Our tool is implemented with Python 3 and uses the open source PULP package which integrates a free solver to solve the above optimization problem. Our preliminary results are very encouraging.
Currently we are working on the resolution of the other chloroplast genomes patterns as directed repeats and combinations of dispersed repeats. In order to compare with the state of the art we intend to solve the benchmarks proposed in [3,4] and hope to be able to present the results on the BiATA 2021 meeting.

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Petabase-scale sequence alignment catalyses viral discovery

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Public databases contain a planetary collection of nucleic acid sequences, but their systematic exploration has been inhibited by a lack of efficient methods for searching this corpus, now exceeding multiple petabases and growing exponentially.
We developed a cloud computing infrastructure, Serratus, to enable ultra-high throughput sequence alignment at the petabase scale (1015 bases). To expand the boundaries of known RNA viruses we searched 5.7 million biologically diverse sequencing libraries (10.2 petabases) for the viral hallmark gene, RNA dependent RNA polymerase. We identify in excess of 100,000 novel RNA viruses and expand the number of known species by roughly an order of magnitude. Among these viruses we characterise novel coronaviruses that are encoded on segmented genomes.
Furthermore, we vastly expand the viruses related to Hepatitis Delta Virus, including some truly enigmatic RNA elements which hint at the early origins of this pathogen.
To catalyse a new era of viral discovery, we established a free and comprehensive database of these data and tools available at (www.serratus.io). Expanding the known sequence diversity of viruses can reveal the evolutionary origins of emerging pathogens and improve pathogen surveillance for the anticipation and mitigation of future pandemics.

BinSPreader: refine binning results for fuller MAG reconstruction

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Despite the recent advances in high-throughput sequencing, analysis of the metagenome of the microbial population still remains a challenge. MAGs are usually constructed via a binning process that uses different features of input data in order to cluster contigs that might belong to the same species. This process has some limitations and binners usually discard contigs that are shorter than several kilobases. Therefore, binning of metagenome assemblies can miss a decent fraction of contigs and resulting MAGs oftentimes do not contain important conservative sequences.
In this work we present BinSPreader – a novel binning refiner tool that exploits the assembly graph topology to refine the existing binning and infer contigs belonging to multiple bins. Furthermore, BinSPreader can split input reads in accordance with the resulting binning predicting reads potentially belonging to multiple MAGs.
State-of-the-art binners use different kinds of information to cluster contigs but the complete information about the assembly is provided via not the set of contigs but the assembly graph. Nevertheless, recently developed graph-aware binning refining tools such as GraphBin2 and Binnacle do not use the assembly graph in the usual sense of the term. Instead, they are relying on the so-called scaffold graph that only preserves the connectivity information between different scaffolds and loses information about the multiplicity of edges and contig's edges. In order to utilize this information we suggest using the original assembly graph.
Firstly, we use the paths of scaffolds over the assembly graph to mark corresponding edges with labels via input binning. Then bin labels of edges are propagated to neighbors via so-called noisy multi-label semi-supervised clustering with possible correction and bin assignment of all scaffolds is determined from bin labels of edges contained. The resulting most probable set of bins the scaffold belongs to is inferred from the bin distribution of a scaffold.
This approach can be used not only for scaffolds, but either for reads to enable fuller MAG reconstruction after the reassembly of subsets of input reads. Besides the assembly graph connectivity BinSPreader is capable of using other kinds of genome connectivity information including paired-end reads, Hi-C links, etc. Utilization of these kinds of input data could further improve the results.
We tested BinSPreader on mock metagenomes BMock12 and Zymo, comprising 12 and 10 known species, respectively. Shotgun metagenomic assemblies for both samples were obtained using SPAdes v3.15.2. Initial MAGs were produced using MetaBAT2. Estimated with AMBER v2.0.2. average MAG F1 score in base pairs increased from 0.601 in initial MAGs to 0.695 in the same MAGs which were refined by BinSPreader for BMock12 and from 0.731 to 0.802 for Zymo.
BinSPreader could effectively improve the binning increasing the completeness of the bins without sacrificing the purity and could predict contigs belonging to several MAGs. Read splitting that takes into account possible overlap between MAGs enables fuller MAG reconstruction after reassembly.

A simple tool for positional view of results generated by comparative genomic analysis platforms in prokaryotes

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Background: Genomic characterization is an essential step in understanding the pathogenic potential of a microorganism. In this process, different platforms for comparative biological data analysis have been used. Summarizing findings is often laborious and time-consuming and requires significant data manipulation to better understand the relationships among different results. Here, we present a tool that offers a visual summary of results obtained in comparative analysis platforms used for prokaryote genomes studies.
Methods: A search in the literature was performed to select the tools used for the characterization of prokaryotic genomes, specifically those dedicated to identifying virulence factors and genetic mobile elements. To build the client interface, we use Electron Js (https://www.electronjs.org/) using JavaScript, HTML, and CSS. In the contents of each output file of the analysis tools, data position is collected, organized, and written in JavaScript Object Notation (JSON) format in order to guide the construction of the image from these positional references. Creating the integrative image was greatly facilitated by JavaScript D3.js - Data-Driven Documents - a library that enables the manipulation of visual elements using web standards. The tool allows the customization of image attributes through an editing interface. The free open access and the adoption of user-friendly interfaces guided the whole process of product development and design.
Results: The platforms selected included the following tools: (i) ResFinder, (ii) VirulenceFinder, (iii) PlasmidFinder, (iv) Phaster, CRISPRCasFinder, (vi) IslandViewer 4. The user can submit the file's results (.csv and tsv) from the analysis of bacterial genomes and access a graphic representation correlating the submitted results in a single circle image, thus speeding and improving the interpretation and understanding of these data. Our image tool was conceived as a desktop application, which works with the most widely used operating systems.
Conclusions: The tool allows the construction of graphical images focusing on the positioning presentation of resistance and virulence genes, plasmids, phages, CRISPRs, and genomic islands in the prokaryotic genomes investigated. With this tool, we intend to significantly improve the work of researchers who face these difficulties.

S3finder (v. 0.2.1) makes finding taxon-specific sequences easier

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The use of PCR-based diagnostic systems is one of the most convenient and popular methods for detecting of pathogens. To carry out accurate species-specific diagnostics by PCR analysis, it is important to select a genome region unique for a taxonomic species and primers that exclude nonspecific amplification. A Linux command line pipeline was developed to perform the search for the taxon-specific sequences. The program is implemented in Python using a special Biopython library designed for solving bioinformatics problems. The pipeline employs BLAST program for the similarity analysis.

The pipeline named S3finder (v. 0.2.1) consists of 3 steps:

1) Data preparation;
2) Search for species-specific areas;
3) Selection of fragments for the construction of PCR primers.

At the first step, all genomes of the target genus are downloaded from the NCBI database and two local databases, so called “positive” and “negative” databases, are formed. The positive database comprises the genomes of the strains (representatives) of the target species. The negative database comprises the strains which should not contain target species-specific sequences. At the same step, the genome of the type strain belonging to the "positive" base is selected and sliced into smaller fragments. To implement the first stage, it is necessary to submit the initial data to the program input. The input data are: taxonomic affiliation of target bacterium (type and genus); target genome INSDC; name of working folder.

At the second step, the search for unique sequences is carried out. The program receives data from the first stage: the file with sliced genome of target type strain and the "negative" database files. It is possible to set some BLAST parameters such as E-value, the number of processor cores and the type of the search algorithm (blastn, megablast, blastp). The pipeline compares the target genome with the negative database sequences and finds out the similar sequences. After that the program creates two files named as “hits” and “no_hits”, which contain analyzed sequences with similarities to the negative database sequences found (“hits”) and not found (“no_hits”). The next part of the second step is the comparison of the “no_hits” file with the “positive” database genomes to reveal the fragments present in all genomes of the “positive” target groups. At the end of the second step, we get a file with target genome fragments which are specific only for the target group (they have similar regions in all the target positive group genomes and do not have homologous sequences on all the negative group genomes).

At the third step, the program works with the file obtained at the second stage concatenating the overlapping sliced fragments found to be taxon-specific. The result of this step is a folder with files each of which contains a region of the target genome that is unique for representatives of the target group. The output files also contain the information about the genomic coordinates of taxon-specific fragments. In future we plan on adding the primer selection function.

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Persephone - a comparative multi-genome browser.

Maxim Troukhan (Persephone Software)

A multi-genome browser, called "Persephone", is presented. The main advantages of the viewer are its fluid speed, interactivity and the ability to handle large amounts of data in real time.
Persephone provides a framework for storing and visualization of genomic data. The platform includes a database designed for fast data retrieval, an interactive viewer with smooth animation, a command-line data loader application, and various analysis services.
The main characteristic feature of Persephone is its ability to show multiple chromosomes on one screen. This allows interactive exploring of syntenic relations between genomic maps based on sequences or genetic linkage groups. The maps can be aligned by connecting orthologous genes, common markers, or by linking regions with high sequence similarity.
Persephone is a modern way for bioinformaticians to present genomic data to biologists. It has been proven efficient in handling millions of chromosome/scaffold maps with different types of data, including gene models, markers, QTLs, variants, RNA-seq, gene expression, BLAST hits, BAM alignments, etc.
A free genomic portal at https://web.persephonesoft.com hosts a number of popular public genomes, such as human, wheat, bacterial or viral genomes (SARS-CoV-2 and the likes). Besides that, the users can also quickly visualize their own data without loading it to the database, just by dragging and dropping local files or reading them from remote locations via URL.
With growing user base in the industry and academia, Persephone continues to uphold its reputation for stability, power, and versatility.

POSTER SESSION

miRPreM : miRNA Prediction Methodology for model or non-model organism using Oryza coarctata, a wild rice as a case study

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The micro-RNAs (miRNAs), a class of non-coding RNAs, are small endogenous regulatory molecules acting as key regulators in post-transcriptional and translational mechanism of genes by interacting with their respective mRNA targets in eukaryotic organisms, including animals and plants. These miRNAs are strongly emerging as the potential targets for disease control in human and to obtain stress-tolerant transgenic plants either through loss or gain of function of target genes. Hence predicting accurate yet enough miRNAs is one of the top priority in this era of data-driven biology. After evaluating the existing methodologies for micro RNAs (miRNAs), we want to propose some recommendations for miRNA identification and annotation by our Standard miRNA Prediction Methodology (miRPreM). We believes that it is a first ever methodology that can be apply for any organism under study matching the gold standard without compromising the quality but increasing the number of predicted miRNAs in a given set of data for any model or non-model organism. Under miRPreM, we highly recommend to perfrom miRNA prediction with the availability of reference genome of organism under study, sRNA data with atleast two biological replicates, preferred length of 17-28 nt length (especially in plants samples), perfect match over entire length for known miRNAs, atleast 1 reads per million (RPM) support in each of the replicates and fulfilling all standard criteria for the novel miRNAs including the availability of longer precursor as well as star sequence with lowest Minimum Folding Free Energy (MFE) and more complicated secondary structure. But most importantly, the miRNA prediction even with the well-accepted standard tools like miRDeep-P2, miRPREFeR and miRPlant (in case of plants) needs to be finalized only after multiple runs due to the variation observed among results from different runs with the same data or identification of more new miRNAs in the new run after removing predicted miRNAs from the input dataset. With miRPreM we registered double the number of predicted miRNAs in our case study with Oryza coarctata, a wild rice as compared to the traditional approaches. We found significant number of organism specific miRNAs because of this miRPreM pipeline which could be either missed if the study has been done with the present practices by taking closely related organism (here Oryza sativa in our case) as reference genome and concluding after just one run from one tool for novel miRNA prediction. Hence, we have a strong belief that miRPreM can fulfill the need of a comprehensive methodology for miRNA prediction to meet the upcoming demands of enough miRNAs.

Elucidation of TAAR1 receptor significance in bladder cancer by the Gene Ontology analysis

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Background: Bladder cancer accounts for 3% of cancer diagnoses, and it is the sixth most incident neoplasm among men. The disease is frequently chemotherapy-resistant and the number of options for second-line therapy is limited.
Trace amine-associated receptors (TAARs) are present in the different organs and recognizing the numerous exogenous and endogenous amine compounds. TAAR1 is considered as the drug target for psychiatric disorders, but its role outside the nervous system remains unclear. The Cancer Genome Atlas (TCGA) data demonstrate that higher TAAR1 expression correlates to longer survival in patients with urothelial carcinoma, but in the same time higher TAAR1 expression is correlated to worse survival in some other cancers. This study aimed to evaluate bladder TAAR1 expression and estimate TAAR1 functional significance in neoplastic bladder tissues by Gene Ontology (GO) semantic analysis.
Methods: Expression data for muscle-invasive bladder cancer (MIBC, n=414) and adjacent normal tissues (n=16) were extracted from TCGA. Non-muscle invasive bladder cancer (NMIBC, n=73) transcriptomes were received from GSE154261 dataset. All data were CPM-normalized. TAAR1-coexpressed genes were selected by Pearson correlation analysis ($r > 0.7$, $P < 0.05$). GO semantic similarity was calculated by Wang's method in the GOSemSim package, the difference was estimated by Brown-Forsythe and Games-Howell tests. Functional profiles of genes and GO enrichment (identification of the genes functional groups that are over-represented in the gene set) was analyzed by the clusterProfiler package.
Results:
TAAR1 mRNA was found in 17% of NMIBC, 18% of MIBC and 37 % of normal bladder tissue specimens. The number of TAAR1-coexpressed genes was 435, 29 and 34 in normal tissue, MIBC and NMIBC respectively. Functional similarity of TAAR1-coexpressed genes in normal tissues was higher compared to the NMIBS (Ptukey<0.001) or MIBC (Ptukey<0.001), that reflected the cancer-associated dysregulation of pathways in which TAAR1 is engaged. Both in the normal and neoplastic tissues TAAR1-coexpressed genes products participate in the signaling, metabolism and cell interactions. No GO-terms enrichment was found in TAAR1-coexpressed genes in MIBC group. In non-cancerous tissues TAAR1-coexpressed genes were enriched with genes involved in keratinization, squamous-cell differentiation, and cornification. It is known that keratinizing squamous metaplasia is a precancerous condition with the risk of malignancy up to 40%. Therefore, the over-representation of mRNA involved in keratinizing squamous-cell differentiation in tissue adjacent to the tumor may reflect not only processes occurring in normal urothelial cells but also

cancer-related disturbances. In contrast, in NMIBC group TAAR1-coexpressed genes were enriched with those related to lipid and steroids metabolism, including androgen metabolism (Padj< 0.05). Sex steroid hormones signaling is considered to be involved in bladder cancer pathogenesis, so TAAR1 seems to be contributed to carcinogenesis-associated events in the urothelium. Conclusion: TAAR1 mRNA is expressed in the human bladder. Cancer-related disordering of biological processes in which TAAR1 is attended was demonstrated in this study. The impact of TAAR1 in carcinogenesis may be a consequence of its possible participation in steroid metabolism and needs to be investigated further.

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Whole genome sequencing of potato varieties of Russian selection with nanopore technology

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Potatoes (Solanum tuberosum) is the third largest food culture in the world. Studying of resistance genes and searching for sites associated with productive qualities are the most important parts of molecular biological research in agriculture. Whole genome sequencing contributes to our understanding of the functionality of the genes and makes them easier to find. Whole-genome sequencing and the search for structural variants (SV) should contribute to the further development of molecular genetic studies on potato crop.
The tetraploid potato genome requires a relatively high coverage (>40) for nanopore sequencing. Fortunately, the comparatively small size of the genome (approximately 840 Mbp) allows to get the required amount of data for 2-4 runs. Also the fact that the potato genome is sufficiently studied means the de novo assembly is not required.
We sequenced genome of three potato varieties of Ural selection. As a result, around 45,000 structural variants of length from 30 to 15,000 bp and around 160 SVs more then 150 kbp have been identified. Only 12.1% of small and 29.1% of large indels found in all genomes were common for three varieties. No structural variants larger than 30 bp in coding regions of the genome of the studied varieties were found.

The pipeline building the fish mitochondrial genome supermatrix for phylogenetic analysis

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The complete mitochondrial genome sequences have proven to be particularly useful for phylogenetic reconstructions with phylogenetic informativeness inferring phylogeny and dating divergence of even relatively distant nodes. The use of the mitochondrial genome in phylogenetic reconstructions is associated with a number of challenges involving the need for a correct annotation, parsing, and alignment of individual genome fragments, as well as the concatenation them into annotated supermatrix. This paper presents FishMitoPipe, a pipeline for parsing an annotated fish genome (produced by MitoAnnotator resource) for an accurate phylogenetic analysis using all (including the control region and ribosomal fragments) or selected genome fragments. The pipeline allows us to perform a partitioning into annotated fragments of the fish mitochondrial genome, align them, and combine them into an annotated supermatrix. This Pipeline is comparable with similar tools (Phylomito, AnnotationBustR, PhyloSuite) and was developed and applied by the author first to work with the control region of fish mitochondrial DNA. It is written on Shell and uses the pyfaidx package (python) as well as the programs seqtk, MAFFT and phyutility. The pipeline is available through https://github.com/Sturcoal/FishMitoPipe and was tested on the Ubuntu 20.04.2 under the Linux Subsystem for Windows.

Transcriptome analysis of crown gall and spontaneous tumors in radish (Raphanus sativus L.) inbred lines

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Studying the aberrations of normal developmental processes like plant tumor initiation may help to gain new insights into systemic control mechanisms in plant cell differentiation and proliferation. Plant tumors can be developed using several mechanisms: either caused by agrobacterium (like in the case of crown gall, caused by Agrobacterium tumefaciens) or be developed without any pathogen infection in plants with a genotype of mutants or interspecific hybrids (in the case of spontaneous tumors). The research to date has tended to focus on the first group of tumors rather than on the second.

This research examines the changes in gene expression at the early stages of A. tumefaciens-induced and spontaneous tumor development in radish seedlings. In our studies, we used two closely related radish inbred lines contrasting in the ability to form different types of tumors. In this work, we analyzed the differential gene expression in the crown gall versus lateral roots as well as spontaneous tumors versus lateral roots of radish using the RNA-seq method and qPCR analysis. There were significant changes in the expression levels for numerous genes involved in different metabolic and developmental processes, and Gene Ontology analysis indicated many biological processes probably involved in the crown gall development in radish. For example, a striking result of our research was the upregulation of genes related to photomorphogenesis and downregulation of senescence-associated genes in the young crown gall tumors. Differences in gene expression levels between analyzed radish lines have also included upregulation of pathways related to chromatin remodeling and protein targeting to the chloroplast in the tumorous line along with downregulation of pathways of chaperone-mediated complex assembly and anthocyanin biosynthesis.

As for spontaneous tumors, data were obtained indicating the increased expression of genes associated with cell division and growth (especially genes regulating G2-M transition and cytokinesis) comparing to the spontaneous tumor and non-tumor lateral root tissues. Among genes downregulated in the spontaneous tumor tissue, genes participating in the response to stress and wounding, mainly involved in the biosynthesis of jasmonic acid and glucosinolates, were enriched. This study has raised important questions about the mechanisms of crown gall and spontaneous tumor development in higher plants.

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Computational Inference of Metabolic Pathways and Phenotypes in Metagenome Assembled Genomes

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Metagenomic sequencing allows to study microbial communities, human microbiota, based on 16S rRNA gene amplicon data. We've developed a new approach, that allows to infer metabolic capabilities of community, e.g. nutrient requirements, utilization, and production capabilities. The approach is based on integration of binary encoded metabolic phenotypes inferred from expert biocuration and in silico genomic reconstruction across a large collection of microbial genomes. Binary phenotypes, where 1 stands for the presence of a particular metabolic capability and 0 stands for its absence, were obtained by mcSEED-based metabolic reconstruction of pathways using their complete genomes. A Binary Phenotype Matrix (BPM) developed in our previous studies, captures utilization and/or production of >100 metabolites including carbon sources (carbohydrates, amino acids), essential nutrients (vitamins, amino acids), fermentation end-products (e.g. short-chain fatty acids) spanning >2,600 reference genomes of Human Gut Microbiota (HGM). In other studies, we showed examples of usability of BPM for predictive functional characterization of metagenomic datasets.
The number of isolated and sequenced HGM microorganisms with available genomes is rapidly growing. Uncultured bacteria with Metagenome Assembled Genomes (MAGs) substantially expand phylogenetic diversity of the collective repertoire of known HGM species. Here we developed and tested a computational pipeline, Phenotype Predictor, for automated propagation of curated metabolic phenotypes from the reference BPM over new microbial genomes from represented phylogenetic groups. First, we built a rule-based (RB) phenotype assignment algorithm using the genomic distribution of orthologs and a set of phenotype rules obtained by the formal Decision Tree approach. Second, we trained several machine learning (ML) models on reference sets of genes and phenotypes comprising >70 metabolic pathways in >2,600 reference HGM genomes. Third, we implemented the neighbor-based (NB) approach that assign a phenotype based on variability of metabolic phenotypes and genes within a group of phylogenetically close neighbors (>90% similarity) from the reference HGM genomic collection. Combination of all three approaches allowed consensus phenotype assignment with >99% accuracy. We report the results of applied them on two recently appeared public sets of >2,300 HGM isolate genomes and >5,000 MAGs to obtain an expanded BPM covering a large phylogenetic space.

Identification and analysis of OR genes in Felids

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Various odor molecules are discovered by olfactory receptors (OR) that are expressed in the olfactory epithelium. OR genes are the biggest family in the mammalian genome. A typical mammalian genome contains 1000-1500 OR genes, that indicates their great importance in the life of many species. Mammalian's sense of smell is exercised to varying degrees in all aspects of life, including finding food, avoiding predators, and social communication. Although the OR gene superfamily accounts for more than 3% of mammalian genes and is well annotated in the human and mouse genomes, it is still not completely known which odoriferous substances bind to which receptors and how this process translates into the recognition of a particular odor.
Because OR genes have a similar structure and most of them do not have introns, the homology search method using already known OR genes is an effective approach. In our work, we used Python to implement modification of the algorithm for identification of OR genes and pseudogenes in mammal genomes, described by Y. Niimura (Identification of Olfactory Receptor Genes from Mammalian Genome Sequences). We have used human's, chimpanzee's, dog's and mouse's OR genes as known genes. Also, we added a step for classification of genes and pseudogenes into specific subfamilies. The algorithm was tested on 3 annotated genomes (cat, cheetah, rat) and showed good results.
Subsequently, the results for currently available genomes of the feline family were obtained and analyzed. Despite the strong kinship, various representatives of the feline family differ in their habitat, family structure, way of obtaining food, lifestyle, and the purpose of our work is to find a correlation between phenotypic traits and the diversity of the OR gene repertoire in felines.
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Using deep learning methods to automatically measure the angle of scoliosis: Bridging the gap between theory and clinical application.

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Scoliosis is one of the most common spinal deformations. It is considered a rich field for technological applications in all its aspects from pathophysiology to postoperative management. Automatic assessment of scoliosis angle is being studied for many years. Though deep learning (DL) of neural networks (NN) had shown a lot of promising results, the best method is until now not defined. Convolutional neural networks (CNN) show promising results in medical image analysis such as segmentation. A review of 13 studies that used DL to measure cobb angle automatically was done, with the aim of evaluating their clinical applicability. Reviewed studies have similar broad lines in their approaches for measuring Cobb angle. Use of a certain data set to teach a proposed network, to detect and segment vertebrae, certain vertebral landmarks or the whole spine, and then measure the angle of inclination automatically. Results were structured to represent certain features of the proposed method and compare them. Those are (type of NN, measurement principle, number of data set and the results). All reviewed studies used networks that were based on CNN with different compositions. The availability of many types of network and the possibility to change their architecture has made the choice of a gold standard method difficult. Spinal isolation and vertebral detection are common initial steps for the automatic measurement methods, which can be divided roughly into two categories: segmentation-based (use vertebral segmentation for angle calculation) and direct estimation methods (without vertebral segmentation). Each method has its limitations. Accurate segmentation of vertebrae is difficult owing to an unclear vertebral boundary in the radiographs. Also, small errors in landmarking can cause serious errors in angle measurement. Regarding the data set, a major drawback in all the studies was its limited number, and also it did not represent all severity ranges of scoliosis equally. Another common disadvantage was the method of evaluation and representation of their results. Direct quantitative comparison was difficult due to differences in the nature of the reported results. Evaluation of such results by mean, for example, which has been seen in most of the studies, cannot be an accurate representative value for all the grades of scoliosis, especially when the training does not involve all the severity ranges equally.
Conclusions: There is a non-neglectable gap between the theoretical method and its clinical application no matter how effective this method is.
It is important to involve a multidisciplinary team of both technical and medical specialists in all the steps of the study, from defining the objectives to the final evaluation process.
To evaluate a certain method, it is more accurate and clinically significant to represent each grade of scoliosis separately, rather than representing the whole data set, especially when the training does not involve all the severity ranges equally. Otherwise, the results will be misleading.
Prospective, comparative or case control like type of studies may be more beneficial to the active and successful integration of any new method into clinical practice.

Genome-wide detection of positive selection in new African cheetah assembly

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Genome-wide scans for genes that have been targeted by natural selection can provide insight into the dynamics of genome evolution, the genetic basis of differences between species, explain adaptations to local environmental conditions and the functions of individual genes. There are two types of natural selection in biological evolution: positive (Darwinian) selection promotes the spread of beneficial alleles, and negative (or purifying) selection hinders the spread of deleterious alleles. PAML is a package of programs for phylogenetic analyses of DNA and protein sequences using maximum likelihood. The package has a rich repertoire of evolutionary models implemented, which include a model that classifies the different codon sites into different categories: evolving under purifying selection, neutral evolution and positive selection. However, this model is highly sensitive to errors in sequence alignment and therefore alignments have the potential to be highly enriched for false positives.

Efficient algorithms are needed to mitigate false positives. One of such algorithms is extensive filtering of alignments using multiple filtering techniques: aligning orthologous genes with the PRANK algorithm, masking of inconsistent residues using GUIDANCE2, further masking for inconsistent regions using two different schemes, like select conserved blocks of sequence with Gblocks or a sliding window approach SWAMP to mask regions of the alignment with excessive amino acid changes.

We have implemented this algorithm in the pipeline, making it possible to find genes under positive selection from genome data. All pipeline steps are accompanied by logging, automatic format conversion, sorting into different species groups of interest, construction of auxiliary files, calculation of significance using the likelihood ratio test, Bonferroni's and false discovery rate corrections, and summarizing results in tables. Pipeline was able to replicate and clarify results from previous study of African cheetah genome. This work was supported by RFBR project number 20-34-70055.

Design and Implementation of Telegram Bots for Biomedical Researchers Support

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With mobile devices in hand, any person can find answers to almost any life question in seconds. And one of the most helpful features in mobile devices is messenger. It becomes the main tool to communicate with each other, to share information and to interact. One of the most useful and technologically advanced messengers is the Telegram. Latest additional feature in Telegram is Telegram bots. At the moment, Telegram bots are widely used for various purposes.

The goal of the project was to build a Telegram bot for simple and intuitive work with the NCBI database, especially with the PubMed database. The developed Bot is aimed at assisting in obtaining information about articles presented in PubMed. Researchers can easily get all the necessary information about articles by sending a keyword or a set of keywords or an author name to the Bot. Moreover, sending the accession number as a request, the user can get a fasta file with desired sequence from nucleotide or protein NCBI database. Also, by sending a document with several sequences in fasta format, the user will receive a document with a built phylogenetic tree, based on the sequence's alignment.

The Bio-Telegram channel of the noticeable Telegram messenger was developed using a high-level implementation library of Telegram’s Bot Application Programming Interface, The Entrez Global Query Cross-Database Search System and The Python Programming Language.

The developed product will reduce the time spent on finding the necessary information for doctors, biomedical researchers, medical students and general public.

Pathway reconstruction and predicting profiling of amino acid metabolism in human gut microbiome using subsystem approach

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The human gut microbiota (HGM) is a complex and versatile community of microorganisms that have a major impact on health and disease by participating in fermentation of dietary polysaccharides/proteins, synthesis of vitamins and short-chain fatty acids, enzymatic transformation of endogenous metabolites. Amino acids are major macronutrients essential for building proteins and synthesis of key primary and secondary metabolites, both in humans and bacteria. Many representatives of HGM community are not able to synthesize some of essential amino acids (auxotrophs), while others show biosynthetic capabilities (prototrophs). Also, degradative capabilities of HGM are just as important for human body since degradative intermediates could serve as bioactive regulators and determine the host metabolic state. Amino acid exchange between auxotrophs and prototrophs can be an important factor shaping microbial community structure.
We analyzed over 2,800 genomes representing 830 cultured HGM species with the aim to identify genes involved in biosynthesis, uptake, and degradation of proteinogenic amino acids and reconstruct corresponding metabolic pathways. Moreover, genome context analysis of incomplete pathway variants allowed us to identify new non-orthologous enzyme displacements in amino acid biosynthetic pathways. We further classified the studied organisms with respect to their pathway variants and infer their prototrophic vs auxotrophic phenotypes (for biosynthetic pathways) and degrader vs non-degrader phenotypes (for catabolic pathways). A cross-species comparison was applied to assess the extent of conservation of amino acid phenotypes at distinct taxonomic levels. Based on in silico predictions we developed phenotype rules and assigned simplified binary (“1” or “0”) phenotypes describing corresponding amino acid prototrophy/auxotrophy in these reference genomes. The obtained binary phenotype matrix (BPM) was used to assess microbiome-wide amino acid biosynthetic and degradative capabilities. We applied the BPM-based predictive phenotype profiling for comparative analysis of human stool samples from three large studies with available 16S rRNA gene sequencing data. For each metagenomic sample and each phenotype, we computed a Community Phenotype Index (CPI) representing the expected fraction of cells possessing a particular phenotype. This in silico analysis of the American Gut project, UK twins and Hadza datasets revealed common trends for certain amino acid biosynthetic capabilities, such as the highest mean CPI values for Lys, Try, Phe, Ser, Met, Pro and Thr (CPI>95%), the intermediate CPI values for Arg, His, and branched chain amino acids (85-95%), and lowest CPIs for Trp (65-75%). We also analyzed the frequency of amino acid biosynthesis genes in whole genome sequenicing (WGS) metagenomic dataset from the TEDDY study containing >1000 fecal samples from infants and 1-3 years old children. The obtained TMM-normalized counts for essential functional roles (signature genes) were summed up for in each amino acid biosynthetic pathway and thus obtained individual

pathway abundance values were compared to corresponding amino acid biosynthesis pathway CPI values calculated using Kraken2-derived taxonomic profiles for the same set of WGS samples. The results of this study and the established technology of phenotype profiling are useful for prediction of overall metabolic properties, interactions and responses of HGM microbiomes as a function of dietary variations, dysbiosis and other perturbations.

Comparative whole-genome analysis of Puma lineage within Felidae

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The Felidae family comprises 41 species which are naturally distributed in almost every area of the world except Australia, New Zealand, New Guinea, and other insular regions, where they have been introduced by humans. Felidae are grouped into 8 lineages based on studies of morphology and genetics. Feline species are diverse with significant variation in size from big cats as Panthera tigris to small cats like Felis nigripes. The karyotypes of Felidae species are predominant diploid number of chromosomes 2n= 38 with alternative chromosome number 2n = 36 for some species. With rapid progress in sequencing technologies whole-genome assemblies are available for 17 Felidae species. In this study we reveal the first whole genome assembly of Puma yagouaroundi. Jaguarundi together with cheetah (Acinonyx jubatus) and mountain lion (Puma concolor) belong to the Puma lineage. Members of the Puma lineage demonstrate unique adaptations to lifestyle, diet and different morphology. Jaguarundi is less studied species in the Puma lineage compared to puma and cheetah, whose genome have been reported previously. Two genome assemblies of cheetah were obtained using by different sequencing technologies, additionally with several specimens sequencing. A study of genomic consequences of inbreeding in mountain lion was performed with the initial genome assembly along with the sequencing data of several other puma individuals. We obtained a high-quality jaguarundi genome assembly using the 10X Genomics linked-read sequencing technology with a 53-fold depth of coverage. The assembly has a total length of 2.47 Gb and is composed of 10,947 scaffolds with an N50 of 49.3 Mb and an L50 of 15 scaffolds. Comparative analysis of genome assembly statistics for the three species indicates that major parts of jaguarundi chromosomes are represented by a few scaffolds, such as for cheetah and puma. Compared to the genome assemblies of cheetah and jaguarundi, the puma genome assembly is more continuous, contains less scaffolds with a greater length, but it has the greater number of gaps. The evaluation of the assembly completeness using universal single-copy orthologs (BUSCOs) indicates that most of the orthologs are present in a single copy in the three assemblies, and only a small number of BUSCOs are duplicated, fragmented, or missing. Comparison of the repetitive genome regions shows no significant difference between three species. Repetitive elements constituted around 40% of the jaguarundi, cheetah and puma genome assemblies, which is also close to 42.8% of repetitive elements identified in the domestic cat assembly. Analysis of heterozygosity that was estimated as the number of heterozygous SNVs in relation to genome

assembly size confirms lower genetic diversity of the cheetah among the Puma linage species. In conclusion, the Puma lineage is the first monophyletic group in the Felidae family with every species represented by a high-level whole-genome assembly. Genome comparisons between closely related species provide insights into mammalian evolution and adaptations and facilitate identification of structural variations, rearrangements, genomics repeats as well as improving gene annotations in the Felidae family. This work was supported by RFBR project number 20-34-70055.

Transcriptome analysis of Medicago truncatula calli with MtWOX2 overexpression

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Somatic embryogenesis (SE) in one of the plant regeneration pathways. This process is very similar to zygotic embryogenesis, but embryos develop not from zygote, but from vegetative tissues. This process is widely used in biotechnology for plant transformation and propagation. Somatic embryos can derive directly from the vegetative tissues or through the formation of callus. The search for SE stimulators is very important for plant biotechnology.
Several genes were reported to be the regulators of this process, among them the WOX (WUSCHEL-RELATED HOMEOBOX) family members are presented. These genes encode homeodomain-containing transcription factors, participating in different developmental processes.
WOX2 is known for its participation in zygotic embryogenesis. It is expressed in the zygote and later in the apical domain of the embryo. We study the role of this gene in the somatic embryogenesis. Overexpression of MtWOX2 in some cases leads to the development of embryogenic calli with increased size. We performed transcriptome analysis of Medicago truncatula calli with overexpression of this gene compared to the calli overexpressing GUS.
It was shown that MtWOX2 overexpression led to the changes in expression levels of genes, enriched with several GO pathways, including groups related to oxidative stress and ROS formation, response to toxic substance and auxin-activated signaling pathway. Among differentially expressed genes there are members of several TF families, e.g. MADS-box, BHLH, MYB, bZIP and others. These genes may regulate the embryogenic callus development. Together, these results can be used for the search of new morphogenic regulators applicable for plant transformation.
The research was supported by RFBR grant 20-016-00124.

Draft genome assembly of the early-branching dinoflagellate Oxyrrhis marina

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Dinoflagellates are microbial eukaryotes with one of the most outstanding nuclear configurations. They have permanently condensed chromosomes lacking the typical nucleosome organization. In terms of the genome, they possess multi-gigabase genomes made up of repetitions and hence largely avoided for sequencing projects. So far, most sequenced dinoflagellates are restricted to the Symbiodiniaceae family, a symbiotic lineage with small genome sizes. Despite this effort, several questions regarding nuclear biology and the evolution of this group remain unanswered, and lesser-explored lineages may provide some insight into these questions. Oxyrrhis marina is a free-living, heterotrophic, well-established model species for dinoflagellates with a large genome size, which is more representative of the genomic expansions that characterize this important eukaryotic group. A detailed exploration of this early diverged dinoflagellate may help to understand the origin of dinoflagellate peculiarities and diversification. Here we present preliminary results from the genome sequencing of the ancestral branching dinoflagellate O. marina. We used a combination of Illumina paired-end and Pacbio CLR sequencing to generate a hybrid assembly. We generated 371 and 9 million of the Illumina and Pacbio CLR reads, respectively. We combined multiple long-read assemblers to obtain a 0.21 Gb assembly and achieve 22% of completeness. We were able to identify a significant fraction of duplicated genes as well as overrepresented satellite repeats. Additionally, the mitochondrial genome was consistently identified in several contigs. This draft genome assembly encourages large genomes explorations and promises to reveal some early genomic features of the dinoflagellate lineage.

Applications of coarse grained enhanced molecular dynamics for antibody structure modelling

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Antibodies are immune system proteins that recognize the surfaces of foreign molecules (antigens) for subsequent elimination from the organism during an adaptive immune response. Continued exploitation of antibodies for therapeutic purposes relies on more efficient ways to develop these molecules. Antibody structures have the potential to be useful during drug development, allowing the implementation of rational design procedures. The most challenging part of the antibody structure prediction is to model the H3 loop, which in addition is often the most important region in an antibody's binding site. The majority of H3 loop prediction approaches are based on computationally expensive decoy generation. The biggest challenge in such ab initio modelling remains selection of best loop models among those generated [1].
The use of coarse grained (CG) models in a variety of molecular modelling techniques has proven to be a valuable tool to probe the time and length scales of systems beyond what is achievable with traditional all atom (AA) models. The recently released CG forcefield Martini3 [2] with an expanded ability to include specific interactions such as hydrogen bonding and electronic polarizability may have a practical use for antibody structure modelling.
We aimed to demonstrate the potential application of CG molecular dynamics (MD) for the specific task of H3 loop conformation sampling. We used pairs of apo- and antigen-bound antibody Fab-fragment structures that have differing H3 loop conformations. The molecular dynamics simulations based on CG forcefield Martini3 were set up starting from an antigen-bound structure with antigen removed. In order to enhance the sampling of the H3 loop conformational space we used two approaches: Hamiltonian replica-exchange (HREX) MD with the H3 loop assigned to be the hot region, and Metadynamics applied to H3 loop dRMSD. The overall Fabs' structure remained stable along the trajectories both in HREX replicas and Metadynamics simulations. The H3 loop region was observed in conformations close to the unbound antibody crystal structures, thus proving the CG MD suitable for antibody modelling.
The reported study was funded by RFBR, project number 19-34-51043, and supported by Russian Science Foundation Grant 21-74-20113. The research is carried out using the equipment of the shared research facilities of HPC computing resources at Lomonosov Moscow State University [3].

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Discovering orthonectids plasmodium-specific genes (Bilateria: Orthonectida)

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Orthonectids are peculiar parasites of marine invertebrates. For a long time, orthonectids were considered evolutionary primitive due to their simple morphology, but recent molecular studies placed them among advanced segmented worms - annelids. Orthonectids are highly divergent from their relatives. They underwent dramatic reduction and lack typical annelid features. The key difference is their lifestyle: orthonectids are endoparasites with a complex life cycle alternating between asexual parasitic and sexual free-living generations, which does not occur in other annelids. Their parasitic stage, the plasmodium, is not found in other bilaterian animals.
The plasmodium is a trophic single-celled multinucleated shapeless organism that inhabits hosts tissues and penetrates the host body. The next, sexual free-living generation develops inside the plasmodium cytoplasm from special cells. Sexual orthonectids leave the host, copulate in water, then a newborn larva infects a new host and turns into a plasmodium again.
Orthonectids dramatic loss of complexity and a unique life cycle is the only case among the large Annelida group, but the origin of orthonectids parasitism is still unknown. The main adaptation to parasitism, the plasmodium, remains underexplored. Discovering genes expressed explicitly in the plasmodium is an essential step towards revealing the molecular processes behind the orthonectids' parasitic stage. It will help to explore orthonectids adaptations to a parasitic lifestyle.
We worked with two close-related orthonectid species, Intoshia linei and Intoshia variabili. The study on a particular life cycle stage implies the use of RNA sequencing. Working on a plasmodium is challenging since it is impossible to separate plasmodium from host tissues. We sequenced the entire infected hosts drawing on the experience of the dual-seq approach. We discarded host contamination by mapping reads to genomes of orthonectids and host-related species. Apart from the plasmodium, the infected host contains developing sexual orthonectids. We removed their protein evidence by aligning proteins to the transcripts of the sexual stage emitted from the host to the water. We characterized the localization of the resulting plasmodium-specific proteins, the presence of signal peptides and conserved domains and annotated them with Gene Ontology.
The comparison between hypothetical proteins of parasitic and free-living stage resulted in hundreds of plasmodium-specific proteins. Obtained proteins are principal for orthonectids adaptation to parasitism. Functional annotation revealed many proteins involved in the plasmodium defence against host immunity, communication with the host, host nutrients uptake, developing and growth inside the host and regulation of life cycle alterations. These processes have never been described so far, and the effectors remained unknown until now. Many revealed proteins are known effector molecules of other endoparasites indicating the orthonectids adaptations to parasitism evolved convergently. Our findings are essential for understanding the molecular basis of orthonectids parasitism.
We used the equipment of the Bio-Bank Center of St. Petersburg State University and ICG SB RAS Bioinformatics Core Facility. This research was supported by the Russian Science Foundation grant № 19-74-10013.

Evolutionary conservatism of amyloidogenic properties of nucleoporins with FG repeats

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Nowadays, the number of functional amyloids is increasing. Many examples were found in prokaryotes and eukaryotes. Moreover, there are conservative functional amyloids among different organisms. In these cases, protein aggregation seems to be beneficial for an organism. Therefore, this property might be conservative for orthologous proteins. At least one known example supports this hypothesis. The amyloid aggregates of the CPEB protein were suggested to play an important role in the long-term memory formation in Aplysia californica, Drosophila melanogaster, and Mus musculus. Amyloid properties were shown for several structural components of nuclear pore complexes (nucleoporins). It has been demonstrated that yeast nucleoporins Nsp1 and Nup100 can form amyloid-like aggregates. In this work we have investigated the amyloidogenic properties of FG nucleoporins among various taxonomic groups. Sequences of Nsp1, Nup1, Nup2, Nup42, Nup49, Nup50, Nup54, Nup57, Nup58, Nup60, Nup100, Nup153 and Nup159 datasets of orthologs were obtained from EggNOG database for the Opisthokonta group. Orthologs lists were manually curated and then the amyloidogenic properties were analyzed with ArchCandy and IUPRed tools. The numbers of potential amyloids (PA) then were summarized for different taxonomic groups: kingdom, phylum and class. According to the analysis, we established that almost all analyzed nucleoporins are potential amyloids. The mean value of PA for different phylums of analyzed proteins varies in the range from 0.717 to 0.99.

Then we searched for regions with conservative amyloidogenic properties. Nucleoporins' sequences were aligned with the MUSCLE algorithm, and the ArchCandy scores were mapped on the alignment. Then the following parameters were estimated for each position: amino acid conservatism, rate of amyloidogenic sequences and the proportion of gaps. Finally, domains with known 3D structure were mapped on the alignment. A certain region was assumed to possess conservative amyloidogenic properties if at least 50% of corresponding sequences were amyloidogenic and located outside the structured part of the protein. We found that FG repeats of Nsp1, Nup49, Nup57 and Nup58, forming hydrogel in nuclear pore complex, contains regions with conservative amyloidogenic properties. This allowed us to suppose that such conservative amyloidogenic properties may play an important role in the nuclear-cytoplasmic transport and formation of the barrier between nucleus and cytoplasm.

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Presumable bacterial symbionts of marine sponges from the White Sea

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Sponges are complex ecosystems that include symbiotic microbial communities. These communities differ in taxonomic composition from those of the surrounding seawater. We investigated bacterial diversity of three marine sponges Isodictya palmata, Halichondria panicea, Halichondria sitiens, and surrounding seawater from the Kandalaksha Bay, the White Sea in august 2016 and 2018. The bacterial fraction was enriched by differential centrifugation. The V3-V4 region of 16S rRNA from total DNA was amplified and sequenced on Illumina MiSeq. Sequences were analyzed with the DADA2, and obtained ASVs were clustered in OTUs. Also, we performed shotgun metagenomic sequencing of the samples and generated metagenome-assembled genomes (MAGs) using MaxBin2, MetaBat2, and CONCOCT. Quality control and taxonomy annotation of MAGs were performed with CheckM and The Genome Taxonomy Database (GTDB). For annotation of proteins, we used BlastKOALA and WebMGA.

We found that the metagenome of each sponge species was dominated by several sponge-specific OTUs found in both 2016 and 2018 samples. These OTUs had low or zero abundance in surrounding marine water and likely belonged to presumable bacterial symbionts of the marine sponges. From metagenome assemblies, we managed to recover MAGs corresponding to the most abundant sponge-associated OTUs. The MAG from H. panicea was classified as Amylibacter and was similar to the genome assembly of a symbiont previously identified for H. panicea from Icelandic waters (Knobloch et al., 2019). Distinct sponge-associated MAGs of I. palmata and H. sitiens belonged to Gammaproteobacteria and have not been previously described. The MAG from I. palmata was related to family AqS2 (order AqS2), while the MAG from H. sitiens was assigned to family LS-SOB (order UBA10353). Previously, representatives of the families have been found among sponge-associated bacteria (Gauthier et al.,2016; Tian et al., 2016). Functional annotation of proteins predicted in assembled MAGs with KEGG, COG, and PFAM databases revealed the presence of vitamin B12 biosynthesis pathway in symbionts of H. panicea and H. sitiens. Vitamin B12 biosynthesis has earlier been demonstrated as a symbiotic feature in other sponge-associated microbes (Thomas et al., 2010).

Conservative blocks in C-terminal regions of 3-D Cry toxins exhibit amyloidogenic properties

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A spore-forming bacterium, Bacillus thuringiensis, is a source of efficient and specific insecticidal moieties primarily produced during the sporulation phase. Of these virulence factors, the most abundant and frequently used class is represented by the so-called 3-D Cry toxins. While these toxins share a rather conservative structure, namely, bearing three domains exerting pesticidal activity, they also possess flanking sequences. Both N- and C-terminal regions are processed when the toxin enters the host gut; however, the latter has been demonstrated to determine the formation of crystals. Nevertheless, the mechanism governing crystallization as well as the diversity of flanking regions are yet to be elucidated. On that account, we summarized conservative blocks within these terminal parts of Cry toxin sequences and examined them for amyloidogenic characteristics possibly modulating the crystallization process. We started with retrieving Cry toxin sequences from all publicly available data sources, including the Genbank database, Bt genome assemblies, and the Bt Nomenclature database. Flanking regions were obtained using domain coordinates provided by CryProcessor, a tool specifically designed for Cry sequence search. Given the extensive diversity of sequences, we developed a novel method for revealing conservative blocks reflecting subdomains based on building k-mer profile with subsequent hierarchical clustering followed by reconstruction of cluster chains, which were then deduplicated using CD-HIT utility. As a result, 204 and 100 blocks were reconstructed for C- and N-terminal sequences, respectively. Noteworthy, C-terminal regions fell into two clusters based on their lengths. The frequency analysis revealed a correlation between the number of blocks and the total sequence length for C-terminal regions, which corroborated length-based clustering: longer sequences possessed 3 to 5 blocks on average, while short ones carried 1-2 blocks. However, the N-terminal region did not display such a pattern. For each unique block, the overall abundance among toxins and the number of amyloidogenic sites predicted by SARP and Waltz algorithms were summarized. While in the N-terminal region, amyloidogenic sites were only sporadically detected by Walts utility, C-terminal sequences possessed two conservative blocks found in more than half of toxins analyzed with frequent amyloidogenic sites revealed by two algorithms. The results obtained demonstrate an immense structural diversity of the C-terminal region of 3-D Cry toxins in terms of the overall region length, number, and variety of subdomains. Moreover, amyloidogenic sites found in conservative blocks may indicate that the C-terminal region can modulate crystallization through local protein aggregation in these sites.
This study was supported by the Russian Science Foundation (20-76-10044).

Genome assembly of plant viruses important for agriculture in the Czech Republic

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Research group of plant viruses and phytoplasmas, Crop Research Institute, Prague, Czech Republic, is aimed at applied research of viruses and phytoplasmas infecting cultivated crops. High-throughput sequencing (HTS) was used for the study of viruses infecting agriculturally important crops. For the HTS analysis, plants of vegetables, grapevines and fruit trees, exhibiting symptoms of virus disease, were selected. RNA was isolated from infected plants, libraries for the Illumina technology were prepared and sequenced in MiSeq or NovaSeq machines, always for 2x150 nt. In the obtained data, viruses were identified in de novo contigs searched by BLAST against local database of viral reference sequences.
Viral agents responsible for the observed diseases were identified: watermelon mosaic virus (WMV) in the pumpkin (Cucurbita pepo) plants; grapevine virus B (GVB) in the grapevine (Vitis vinifera) plants and Prunus necrotic ringspot virus (PNRSV) in the plum (Prunus domestica) trees. Genomes of the mentioned viruses were assembled de novo using Geneious and SPAdes assemblers. Performance of the two assemblers was compared on the three datasets containing different read numbers and different viruses.
Assembled genome length of WMV was 10027 nt, GVB has 7600 nt. PNRSV has three RNA molecules, their lengths were determined as 3192, 2512 and 1901 nt, respectively.

The work was supported by institutional support MZE-RO0418 from the Ministry of Agriculture of the Czech Republic.

The use of RNA-seq contaminants as a metatranscriptomic data for screening of plant pests and symbionts

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Raw data from transcriptome sequencing could contain up to 30% contaminating reads. They are usually filtered out due to a deteriorating effect on a de-novo assembly quality and subsequent expression analysis. The origin of contaminating reads is either a laboratory introduced contamination or the biologically relevant data, which could be studied using metatranscriptomics approach. The presence of contaminating RNA in an otherwise clean sample could indicate that this RNA is either from a living symbiotic organism or it is somehow recently circulating inside the target organism. Studying the RNA from symbionts can give information of immediate surroundings of the host organism and potentially should be taken into account for gene expression studies since such interaction of host plant with it's symbiome can alter plant's gene expression and mask or mimic expression changes at RNA level and the overall phenotype.

We analyzed the data from our own RNA-seq experiments of Secale cereale (five accessions) and publicly available datasets (60 accessions) from NCBI SRA archive. For the taxonomic classification we used Kraken2 with a full RefSeq database combined with rye genome, rye specific viruses and aphids not yet included in the Refseq. We compared search results obtained by searching quality trimmed raw reads and de-novo assembled transcriptome and found comparable results, which allows us to to use quality trimmed raw reads for the rest of the work as it gives a direct count of reads for each species.

Every accession, even from plants grown in a controlled greenhouse, has a number of contaminating species. Some of them, such as Staphylococcus epidermidis, S. hominis, Cutibacterium acnes etc., are from laboratory contamination, others, such as Homo sapiens, Mus musculus are apparently an interfering bar-coding artifact from samples run in parallel on the same sequencing run, and Methylobacterium sp. is a common contaminant of extraction kits. After removing these species from the list, a number of Aphids, symbiotic fungi, bacteria and viruses present were compared between accessions. The distribution of different species in accessions from the same and different places supports the idea of using available and upcoming sequencing data as a screening for distribution of plant pests, control of new symbiotic organisms and planning on their eradication in the face of the global climate change. This approach seems feasible because a number of reasons: the data is already available and will be even more accessible in future, it can be possible to devise a semi-quantitive analysis, it can be reanalyzed for newly sequenced organisms, the traditional screening using ITSs or 16S rDNA sequencing will produce mostly ITSs of the rye itself and 16S rDNA of it's mitochondrion and chloroplasts, specialized test systems can detect only known few species and are not an option for screening of whole symbiome.

It would be beneficial to add additional information fields to SRA's experiment description including the year and month of material harvest, exact geographical coordinates, growth conditions, including information on used pesticides and information of samples which were run in parallel on the same sequencing run.

Bioinformatic analysis of promoters of Hif's gene isoforms and evaluation of the possibility of their simultaneous transcription under the action of hypoxia

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Introduction. From the point of view of clinical practice, the most common pathology associated with a sharp lack of hypoxia is strokes. The main stage in the practice of emergency care is the "therapeutic window", which lasts up to 6 hours, when it is necessary to carry out thrombolytic therapy or surgery to eliminate the etiological object. Therefore, clinicians conduct a scientific search for the necessary data on the possibilities of diagnosis and relief of such conditions. Accordingly, it is necessary to develop drugs that allow early activation and increase the body's resistance to hypoxia.
The Hif gene and its isoforms are responsible for the development of the adaptive response of the cell to hypoxia. The main isoforms of the gene of interest under consideration are the following species: Hif1a, Hif2 or EPAS-1, Hif3a. In this regard, their activation helps cells and, accordingly, tissues to rebuild under pathological conditions of lack of oxygen. However, this requires an accurate determination of the promoter zones, from which the process of recognition by RNA polymerase and the beginning of transcription will begin with the simultaneous or sequential involvement of other isoforms.
Aim: researching the promoter zones of the Hif gene isoforms and to carry out a comparative analysis of the sequences using the Needleman-Wunsch alignment algorithm.
Materials and methods. To study and determine the promoter zones of the isoforms of the considered Hif gene, the corresponding regions were analyzed in the Human Genome Browser using the “Reference Sequences” annotations. The parameters for taking into account the onset of transcription were the values when the tracks were connected, the level of expression of histone labels, the level of sensitivity to DNase taking into account the number of transcription factors, the degree of expression of Cpg islands, and taking into account the level of CAGE peaks. Isoforms Hif1a and Hif2 / EPAS-1 had similar annotations in RefSeq NGMD, while Hif3a had several versions, for each of which the promoter zones were considered.
Results. For Hif1a, the promoter zone with coordinates chr14 was determined: 62,162,119-62,214,977 with a nucleotide sequence length of 2070 units; Hif2 / EPAS-1 with a promoter located at chr2: 46,524,541-46,613,842 4603 nucleotides in length; for Hif3a, the length of the sequences of the promoter zones was 324, 254, and 289, respectively. The best alignment results by the Needleman-Wunsch algorithm, taking into account the calculation of the sum of substitution penalties, showed sequences in the Hif1a and Hif3a1 promoter zones; Hif1a and Hif3a2; Hif2 / EPAS-1 and Hif3a2. It is theoretically possible to create a drug based on the cascade

isolation of substances that activate the promoters of the Hif gene isoforms in the Hif3a1-Hif1a-Hif3a2-Hif2 / EPAS-1 sequence.
Conclusions. The Hif gene isoform annotations were analyzed for identification of promoter zones in the Human Genome Browser. The obtained sequences were analyzed using the Needleman-Wunsch algorithm for the question of the possibility of creating a promoter compound that would allow their cascade activation, potentiating the body's resistance to hypoxia.

Identical Protein Group content and resequencing statistics as a naive metric of biological assembly quality: An evaluation study and draft tool implementation

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With the advance of next-generation sequencing techniques, repeated sequencing of samples with an existing record of assembly has become a common procedure. Due to their vast diversity and small genome size, prokaryotes are subjected to genome resequencing ever so often, while in eukaryotes repeated sampling is restricted mostly to transcriptomic assays. Its source notwithstanding, each assembly brings along protein-coding sequences, which may be further clustered and stored together based on sequence similarity or identity. This is the case of NCBI Identical Protein Groups (IPG) database in which each cluster is characterized by a total number of occurrences in genome assemblies with source organisms specified. Naturally, given proper taxonomic attribution, each novel assembly is expected to diverge from the available assemblies by a small fraction of protein content, with prominent differences attesting to rapid gene evolution, misassembly, or erroneous species evaluation. Thus, the number of IPG entries found may serve as indirect quality statistics for novel genome assemblies. Here we propose an implementation of this approach for prokaryotic genomics. The average values of total and non-unique ('core') IPG content were estimated for 243 prokaryotic species with a reliable number of high-level assemblies. The respective intervals were then used for quality estimation for assemblies of different completeness levels and taxonomic relevance, including mock ones. The suite of scripts used in this work was gathered into a single user-friendly tool distributed freely together with the compiled species-specific IPG databases. Further expansion of this suit will be aimed to adjust the elaborated protocol to quality assessment of larger eukaryotic genomes as well as transcriptomic assemblies.
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Diagnostics of bivalve transmissible neoplasia in Mytilus trossulus by Illumina amplicon sequencing

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Bivalve transmissible neoplasia (BTN) is a leukemia-like disease that is passed from one individual to another by the physical transfer of cancer cells (neoplastic hemocytes). The neoplastic hemocytes firstly invade the hemolymph of the mollusk and while disease progresses they enter the other tissues and organs. The genotype of neoplastic hemocytes is different from that of the host, so the diseased bivalve represents a genetic chimera - an organism possesses cells with different genotypes. The discovery of the contagious feature of cancer in marine bivalves is fairly recent (Metzger et al, 2016).
The straightforward approach for BTN diagnostics is genotyping of different tissues (traditionally, hemolymph and foot) by three diagnostic loci an intron-spanning region of the nuclear elongation factor 1α gene (EF1α), a fragment of the mitochondrial cytochrome oxidase subunit I (COI), and a fragment spanning 16S rRNA and control region (CR).
While healthy individuals' amplicons represent the identical sequences from hemolymph and foot tissue, the chimeric individuals with BTN have a mixture of cancer and host sequences in their amplicons. The common procedure for resolving this mixture and identification of nucleotide sequences imply molecular cloning and Sanger sequencing of at least 6 - 16 clones in order to establish the whole diversity of variants present in the sample (Metzger et al, 2016, Yonemitsu et al, 2019, Skazina et al, 2021). The NGS techniques could be less cost and effort-consuming than molecular cloning in this case and possibly provide more precise results for detecting the BTN in the early stage of disease when the cancer alleles are present at a very low level.
In this work, we are the first to assess the Illumina amplicon sequencing for diagnostics of BTN in blue mussel Mytilus trossulus.
Two control healthy specimens together with four M. trossulus from the Sea of Japan, Russia, previously diagnosed by molecular cloning as BTN infected were selected for this study. Fragments of EF1α (521 bp), COI (502 bp), and CR (497 bp) were amplified from hemolymph and foot tissues of all specimens with specific primers developed for this study. Three types of amplicon from each individual were pooled together and sequenced in pair-end mode on Illumina MiSeq (2x250) on the basis of the Research Park of the Saint-Petersburg State University. The raw pair-end reads underwent quality assessment, filtering, and trimming by means of FastQC (Andrews, 2010) and Trimmomatic (Bolger et al, 2014). The reads were merged with NGmerge (Gaspar, 2018) and aligned to a reference sequence by the bwa mem algorithm (Li et al, 2009). The analysis of preprocessed reads was performed with the standard pipeline for the dada2 R package (Callahan et al, 2016).
The resulted lists of sequences detected by two approaches - molecular cloning and NGS technique showed good correspondence. Nevertheless, the Illumina amplicon sequencing detected the

presence of additional cancer alleles that were not detected by means of molecular cloning. We suggest the pipeline used in our study as a standard approach for BTN diagnostics.
This study was supported by Russian Science Foundation, Grant Number 19-74-20024.

Phylogenetic Relationships of Ascomycetes Opine Synthases

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Opines are low molecular weight compounds that are synthesized in organisms of different taxa. They owe their name to octopine, first discovered in cephalopods, where it is formed during the anaerobic oxidation of glucose by octopine dehydrogenase through the condensation of pyruvate and L-arginine, followed by reduction to octopine. At the moment, the most famous are opines of agrobacterial origin. These are strain-specific metabolites that are synthesized in the crown galls and hairy roots induced by these bacteria due to the expression of the transferred genes. They are metabolized by the corresponding bacterial strains. Recent studies show that there are a number of examples when agrobacterial opine genes transferred into plant genomes millions years ago retained their integrity and are expressed. It is believed that the products of these genes play a role in the regulation of plant-microbial interactions [1]. Genes for opine synthesis were found not only in plants, but also in fungi. However, genes were annotated as opine synthases in a number of independent studies based only on bioinformatics approaches. Classical biochemical studies confirming the synthesis of opines by fungi have not been carried out, but opine utilization was documented [2]

The aim of this study is to analyze the phylogenetic relationships of the ascomycete opine synthases. Methods of the analysis are described in our previous works [3].

At the first stage of our study, we analyzed the similarity of the amino acid sequences of opine-synthases from various groups of organisms. The analysis included both bacterial and eukaryotic sequences and was limited to octopine/nopaline dehydrogenase family proteins, since they are annotated in the largest spectrum of organisms. Phylogenetic analysis of the sequences of bacteria, ascomycetes fungi, marine invertebrates, naturally transgenic plants, etc showed that there are three clusters of the studied sequences. The first one includes opine synthases of marine coelenterates, worms, and mollusks. The second cluster is formed by sequences of bacteria, naturally transgenic plants, and ascomycetes. The third cluster is a mixed group. The interpretation of it should be done very carefully, since contamination of DNA samples of symbionts is not ruled out here.

In the genomes of fungi of classes Pezizomycetes, Eurotiomycetes; Leotiomycetes; Saccharomycetes; Sordariomycetes; Blastocladiomycetes; Mucoromycetes there are homologues of the genes ags/chs, ocs/vis, mis. An analysis of the phylogenetic relationships shows mosaic arrangement of species containing different opine synthases on the phylogenetic tree. It suggests the role of horizontal transfer in the appearance of these sequences in the genomes of fungi or evolution from a common ancestral sequence by losing part of it. Considering that among fungi there are also many species that enter into symbiotic relationships with both plants and other microbes, then the idea of the role of opines in the regulation of plant-microbe interactions looks even more intriguing and requires close attention.

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Genome-wide Prediction of Transcription Start Site in Conifers

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Despite being an ancient group of plants represented by more than 600 species conifer genomes remain largely understudied, due to their enormous size and highly repetitive nature. In the last years, several mega-genomes of conifer species were sequenced and assembled to the draft state, and although their current annotations are preliminary and limited, nevertheless they provide opportunities for structural and functional analysis. Understanding of regulatory relationships between genome elements requires information on promoter sequences, which are usually located directly upstream or at the 5' end of the transcription start site (TSS). Here we present our efforts to improve the existing genome annotations by predicting TSSs and marking 5’-UTRs in the four relatively recently published species Pinus taeda, Picea glauca, Picea abies, and Larix sibirica.
All gene models retrieved from genomic annotations were aligned against the database of RNA-seq data, including ESTs and TSAs of a corresponding species using hisat2 alignment program and to Refseq plant protein sequences using blastp, to filter out possible pseudogenes and putative predicted coding sequences. Prediction of putative TSSs was performed in the regions of -1000 and +250 bp around the start codon of selected genes, using TSSPlant program. Selection of the best prediction was done based on the distribution of 5’-UTR lengths from the annotations of several model plants. The accuracy of predictions was assessed based on the free energy profile and frequencies of CA and TATA motifs calculated within a sliding window on TSS-centered sequences.

Hybrid de novo genome assembly of the most halotolerant midge Baeotendipes noctivagus

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Studies of the genomes of extremophilic organisms allow us to investigate how these organisms adapted to extreme environmental conditions (complete desiccation due to aridity, high salinity, extreme acidity, low temperatures) via gene adaptations, changes in expression levels, and how the evolution of extremophilic organisms in general occurred. To perform genomic and transcriptomic analysis we need to have high quality reference genome assembly.
In this work, we have sequenced a record holder of halotolerance among insects - Baeotendipes noctivagus, a midge from Chironimidae family (Diptera) that can withstand a wide range of salinity from 20g/L to 300g/L in the larval stage. We performed hybrid de novo genome assembly using genome assembler - Wengan that utilized paired-end short reads (2X250) obtained by Illumina HiSeq 2500 platform and long reads obtained by MinION nanopore sequencing (N50 = 15Kbp). Long reads data obtained from DNA isolated from a pool of individuals to obtain sufficient amount of DNA, while short reads data obtained from DNA isolated from a single larva of the same population. Total coverage by Illumina reads estimated as ~110X in one sequencing run and total coverage by Nanopore reads estimated as ~50X after two sequencing runs, which was sufficient for hybrid assembly. The total length of the resulting assembly was 177.4Mb, N50 = 1.9Mb. Genome assembly consisted of 529 contigs. The largest contig in obtained assembly is almost ~7Mbp. Genome assembly was polished by HyPo software with both types of libraries. BUSCO analysis showed that completeness of the assembly after the polishing step using the diptera_odb10 database was 94.0%[S:92.9%,D:1.1%],F:1.1%,M:4.9%,n:3285, which is very good result that allow to perform comparative genomic analysis.
Additionally, we performed poly(A) RNA sequencing in pair-end mode (2X100) using Illumina HiSeq 2500 platform from three populations of Baeotendipes noctivagus (one of them used as reference genome), where all three populations of Baeotendipes noctivagus lived in different salinity regimes (20g/L, 80g/L and 300 g/L). RNA-seq data from the population that lived in medium salinity was used for genome annotation that was performed by the BRAKER2 pipeline. Transcriptomic analysis using edgeR showed differentially expressed genes between these populations.

In this work we showed that third generation sequencing technologies significantly simplify genome assembly of non-model insects. We plan to perform comparative genomics with other halotolerant midges to investigate evolutionary adaptations (similar and distinct pathways). This work has been supported by the Russian Science Foundation (grant №20-44-07002).

HORmon: automated annotation of human centromeres

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Recent advances in long-read sequencing led to rapid progress in centromere assembly and opened a possibility to address the long-standing questions about the architecture and evolution of human centromeres. They also emphasized the need for centromere annotation (partitioning human centromeres into monomers and high-order repeats (HORs)), a prerequisite for follow-up biomedical and evolutionary studies of centromeres. However, even though there was a half-century-long series of studies of centromere architectures, a rigorous centromere annotation method is still lacking. Although the monomer decomposition (transforming a centromere into a monocentromere written in the monomer alphabet) and the HOR decomposition (representing a monocentromere in the alphabet of HORs) are currently viewed as two separate problems, we demonstrate that they should be integrated into a single framework in such a way that HOR (monomer) inference affects monomer (HOR) inference. We thus developed the HORmon algorithm that integrates the monomer/HOR inference and automatically generates the human monomers/HORs that are largely consistent with the previous manual annotations. HORmon complements the classical concept of a HOR by the concept of a monorun-graph that provides a more informative description of the centromere architecture than a HOR.

Computational prediction of biological activities of peptidic natural products

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Peptidic Natural Products (PNPs) are a rich source of antibiotics and other pharmaceuticals. Recent breakthroughs in mass spectrometry (MS) and genome sequencing enabled high-throughput PNP discovery. Emerging computational methods allowed identification of novel PNPs via modification-tolerant database search of MS data [1], de novo MS sequencing [2], and metabologenomics approaches [3, 4]. These methods produce large amounts of in silico predicted PNPs that require further experimental analysis. Creating a robust strategy for selecting the most promising compounds out of many computationally predicted molecules remains an open problem.
In this work, we develop a computational pipeline for predicting biological activities of PNPs based on their chemical structure. The software accepts PNP structures in standard chemical formats such as SMILES or MDL MOL. The input molecules are further converted into a unified form and searched in public chemical databases to retrieve any associated data on the proven biological activity of precisely these or structurally similar compounds. All retrieved information is combined into a single detailed report per PNP and presented to the user. The pipeline is implemented in Python with the use of the RDKit library. The currently only target chemical database is PubChem [5]. The support for other common databases is scheduled for future releases of the pipeline.
We plan to integrate the developing pipeline with the state-of-the-art approaches to PNP discovery [1-4] to complement their output with the tentative biological activity of the identified compounds. We anticipate our pipeline will facilitate the search for new therapeutic agents by providing researchers with biologically relevant information. This data will help to prioritize in silico predicted PNPs for experimental validation and testing.

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Genome and transcriptome assembly in Dasyhelea calycata.

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Salinity is an important abiotic factor, which drives the diversification of animals and plants. It was previously shown that the expression patterns are changed due to salinity change in a wide range of organisms such as fishes, crustaceans, nematodes, and molluscs (Heckwolf et al. 2020, Rohlfing et al. 2020, Gao et al. 2018, Zhao et al., 2016). The osmotic stress is related to the ion channel activity in the excretory epithelial cells. However, some indirect links were also proven, such as the inhibition or activation of an immune response, synthesis of the ROS, and regulation of the lipids metabolism in the liver. The midge Dasyhelea calycata (Ceratopogonidae) is one of the few insects that can adapt to high salinity. Thereby it can be an excellent organism to study adaptation on the molecular level.

To investigate the molecular adaptations to the high salinity changes, the genome of D. calycata was sequenced with Illumina Hiseq 2500 (Paired-end library 250+250 b.p.) and further assembled with SPAdes (v. 3.15.2), MaSurCa (v.3.4.3), and Meraculous (v. 2.0.4). Also, the transcriptome of the midge was made with RNAspades (v. 3.13.0) and Trinity (v. 2.8.4). In order to understand the rate of expression of each gene. The quality of genome and transcriptome assembly was accessed with metrics generated by Quast and by BUSCO analysis. Only 66.6% of complete genes were found in the assembly generated by SPAdes with 254063 contigs and 82% of completed genes with 10825 contigs in MaSurCa. Thus, to improve genome data, we are expected to use nanopore data. Along with long reads, we will be able to conduct further analysis.

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Metataxonomic and functional profiling of oil well microbiome from Moskovtsev field

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Microbial studies in oil and gas fields can determine how the microbiota of oil and gas reservoirs affects the technological processes of oil and gas production. In this work metataxonomic analysis of 179 samples of oil fluid, drilling cuttings and drilling mud was carried out to determine the composition of the microbiota, and the possibility of using the knowledge of the microbiota of wells for practical purposes.

Microbiome composition was examined by sequencing of 16S rRNA V3-V4 regions and sequenced by Illumina MiSeq sequencing platform. Dada2 pipeline was used to infers exact amplicon sequence variants (ASVs). To perform a taxonomic profile of samples we used Silva (v132) database. Functional profiling of obtained amplicon data was performed by the PICRUST2 pipeline to find oil-degrading metabolism pathways by using KEGG, PFAM and MetaCyc databases.
Taxonomic profiling, PCA showed that microbiota samples mainly clustered according to different lithological compositions (argillite, sandstone, siltstone) of the drilling fluid. The most abundant genus in almost all samples are presented by Halomonas, Dietzia, Marinimicrobium, which are shown to be oil-degrading microorganisms and very common organisms in oil fields.
Functional profiling of fluid and drilling cuttings samples showed that the degradation pathway of polycyclic aromatic hydrocarbons was enriched in fluid samples compared to the microbiome of the drilling fluid.
Our results will help in the development of technological applications in oil field exploration using microbiome data.
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Methylation patterns of Rhizobium leguminosarum genome change in terminally differentiated cells

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Bacteria of the Rhizobia group possess a unique ability to form symbiotic relationships with the plants of the Fabaceae family. The or so-called "cross-talk" of molecular signals, plays a crucial role in the formation of said symbioses helping to synchronise processes in the symbionts, resulting in the formation of nitrogen-fixing nodules. During the formation of symbiosis both the organisms undergo drastic changes to their physiological state. The plant forms the symbiotic nodule, while the bacteria terminally differentiate into bacteroids - a nitrogen fixing form, uncapable of free living. Although the multiplication of bacterial DNA is a well-established fact, the process was not investigated in detail previously for pea plant in particular.
Objective of this study was to characterise the genomic composition of the RCAM1026 strain in free-living conditions and in bacteroids to identify the changes occurring during terminal differentiation.
The strain RCAM1026, routinely used for inoculation experiments, was investigated. The free-living cell culture was harvested after one day of growth in 28°C in TY media. The bacteroid DNA was obtained from bacteroids, which were isolated from four weeks after inoculation nodules formed on the roots of Frisson cultivar plants. MinION sequencer (Oxford Nanopore) was used for genome sequencing.
First the genomes of two bacterial forms were compared in order to investigate the possible genomic recombination events. The resulting assembly for the free-living culture consisted of 5 circular replicons, of which the symbiotic plasmid showed two times lower coverage. The bacteroid genome assembly showed no distinct structural variations, but the relative multiplicity of the chromosome was two times that of the rest replicons, meaning that the chromosome is multiplied more during the bacteroid formation. Interestingly, all the plasmids had very similar coverage.
Next the nanopore reads were used to determine the methylation of genome. WG amplification was performed, and the resulting amplicons were sequenced using a MINion sequencer to obtain reads of non-methylated DNA. Megalodon and Nanodisco algorithms were used to investigate the differences in methylation between the free-living culture and bacteroids. Two distinct methylation patterns were found in free-living culture: GANTC and GGCGCC; only one distinct methylation pattern was found in the bacteroid genome: GGCGCC. The GGCGCC site seem to represent a restriction enzyme site, previously not described for R. leguminosarum. The GANTC motif was much less prevalent in bacteroids, than in free-living culture, which can be linked to the terminal differentiation of bacteroids, which do not undergo cell fission and can result in the increase in chromosome copy number. The lack of methylation may be one of the mechanisms switching the bacteria from free-living to differentiated state. Investigation of methylation patterns adjacent to

specific genes in conjunction with transcriptome profiling will shed light at specific genes, regulated with methylation in bacteroids.

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Differences in the metagenome of microbial cellulolytic communities on various substrates

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Stable microbial communities decomposing cellulose on various substrates (straw, litter, sawdust) were isolated on the basis of the BAGS biopreparation. We examined the metagenome of four communities. The 16SU rRNA libraries for bacteria and ITS for fungi were sequenced using Illumina technology. In addition, metagenomic sequences were assembled for each community using Oxford Nanopore. The aim of the work was to find differences between microbial communities metabolizing cellulose on different substrates using, among others, ONT assemblies. Each community was represented by more than a hundred microorganisms. A strict taxonomic binding of the communities to the substrate was shown. Twenty families of glycosidic hydrolases with cellulolytic/hemicellulolytic activity were found in metagenomes. In addition, 10 complete genomes were isolated from the assembly. However, the obtained genomes do not belong to the taxa in which the highest number of glycoside hydrolases was detected. Despite the strong differences in taxonomy between metagenome assemblies and 16S rRNA, no strong differences in the representation of different glycoside hydrolase families were shown in the metagenome.

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Nodule metagenomes of pea symbiotic nodules formed in two different soils is influenced by the plant and the soil.

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Garden pea can form symbiosis with nodule bacteria, as a result of which nodules inhabited by bacteria, which fix atmospheric nitrogen, develop on the roots of plants. In this superorganismal system, the plant genome is “complemented” with bacterial genes, which leads to the appearance of new properties in the system (including the ability to fix nitrogen), which are not characteristic of each of the symbionts separately. The efficiency of nitrogen fixation and the benefit from it for the plant-microbial system depends on both the plant genotype and the bacterial genotype.
This study examines the question of whether a plant is able to select the most beneficial bacterial strains from the soil, or whether the bacterial community plays a leading role in the formation of the “symbiotic microbiome” profile.
In the experiments, plants of lines Frisson, Triumph, 8274, 3358. "Chernevaya Taiga" and sod-podzolic soils were used. Pea plants were inoculated with PBS-based suspensions of soils from two distinct geographic regions. The plants were used to analyse the efficiency of nitrogen fixing using the acetylene-reduction assay and for the isolation of bacteroids. The complete metagenome of the microbial community of bacteroids in nodules was sequenced using the Oxford Nanopore sequencing technology. Assembly was performed using flye (v 2.8), the resulting assemblies were compared to each other, and to a meta-assembly from all the nodules. It was possible to recover up to 20 megabases of genomic data from the sequencing samples (2-3 complete rhizobial genomes), after multiple iterations of removing the inevitable host DNA contamination. Only sequences belonging to Rhizobium leguminosarum species were assembled, proving that the majority of bacterial DNA indeed came from the bacteroids. Comparison of the formed nodule communities showed that, as expected the communities were more dependant on the soil inoculant than on the genotype of the plant. We also found that the “Chernevaya Taiga” soil harbors a diverse array of Rhizobium leguminosarum, capable of forming highly efficient nodules, according to the nitrogen fixation data. Further investigation into the identity of possible host in the native plants is needed.
In this work, for the first time, complete information was obtained about the "symbiogenome", which is the result of complementation of the plant genome with a set of genomes of nodule bacteria. The experimental results support the hypothesis that the success of the symbiotic system as a whole depends on the competitiveness of the strains to a greater extent than on the selectivity of the plant. Consequently, it seems promising to directly create the competitive and effective strains of nodule bacteria, which will be the first to penetrate into the plant and determine the composition of the "symbiogenome" of the plant-microbial system.
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